Main Category	Category	Terms
Disease	OMIM_DISEASE	Tubulointerstitial kidney disease, autosomal dominant, 2
Disease	UP_KW_DISEASE	Tumor suppressor
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Membrane
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoplasm
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Secreted
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cell membrane
Functional Annotations	UP_KW_PTM	Autocatalytic cleavage
Functional Annotations	UP_KW_PTM	Glycoprotein
Functional Annotations	UP_KW_PTM	Lipoprotein
Functional Annotations	UP_KW_PTM	Palmitate
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Disulfide bond
Functional Annotations	UP_SEQ_FEATURE	CARBOHYD:N-linked (GlcNAc...) asparagine
Functional Annotations	UP_SEQ_FEATURE	CARBOHYD:O-linked (GalNAc...) serine
Functional Annotations	UP_SEQ_FEATURE	CARBOHYD:O-linked (GalNAc...) threonine
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Basic and acidic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Low complexity
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Polar residues
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:SEA
Functional Annotations	UP_SEQ_FEATURE	LIPID:S-palmitoyl cysteine

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	MOTIF:Interaction with GRB2
Functional Annotations	UP_SEQ_FEATURE	MOTIF:Interaction with SRC and ESR1
Functional Annotations	UP_SEQ_FEATURE	MOTIF:Required for interaction with AP1S2
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:C->A: S-palmitoylation reduced by 50%. Complete loss of palmitoylation, no effect on endocytosis, recycling inhibited and AP1S1 binding reduced by 30%; when associated with C-1186. Accumulates in intracellular compartments; when associated with C-1186 and N-1203.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:C->A: S-palmitoylation reduced by 50%. Complete loss of palmitoylation, no effect on endocytosis, recycling inhibited, and AP1S1 binding reduced by 30%; when associated with C-1184. Accumulates in intracellular compartments; when associated with C-1184 and N-1203.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:D->A: Greatly reduced formation of isoform 5/isoform Y complex.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:D->E: No effect on formation of isoform 5/ isoform Y complex.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:RRK->AAA: No nuclear targeting of HRG-stimulated MUC1 C-terminal nor JUP/gamma-catenin. No effect on interaction with JUP/gamma-catenin.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:RRK->QQQ: No effect on palmitoylation.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A,D,E,F,G,H,I,K,L,M,N,P,Q,R,V,W,Y: Completely abrogates cleavage.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A: No change in PRKCD- nor GSK3B-mediated phosphorylation.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A: No change in PRKCD-mediated phosphorylation. Loss of GSK3B-mediated phosphorylation. CTNNB1.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->C,T: Almost complete cleavage.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->A: Loss of PRKCD-mediated phosphorylation. Decreased PRKCD binding. No increased binding to CTNNB1 in the presence of autophosphorylated PRKCD. Increases formation of E-cadherin/beta-catenin complex.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->E: No effect on nuclear colocalization of MUC1CT and CTNNB1. No effect on in vitro PDFGR-induced cell invasiveness.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->F: Greatly reduced EGFR- and Src-mediated phosphorylation. No nuclear localization of MUC1CT. Reduced in vitro PDGFR-mediated

Main Category	Category	Terms
		phosphorylation. Decreased Src-binding.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->F: No effect on EGFR-mediated phosphorylation.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->F: No effect on EGFR-mediated phosphorylation. No nuclear colocalization of MUC1CT and CTNNB1.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->F: No effect on EGFR-mediated phosphorylation. No nuclear localization of MUC1CT. Reduced in vitro PDGFR-induced cell invasiveness.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->F: Some reduction in EGFR-mediated phosphorylation.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->N: No effect on endocytosis.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->N: Reduced endocytosis by 30%. Greatly reduced binding to AP1S2 and GRB2. Binding AP1S1 reduced by 25%. Reduced endocytosis by 77%; when associated with N-1243. Accumulates in intracellular compartments; when associated with C-1184 and C-1186.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->N: Reduces binding to AP1S2 by 33%. Greatly reduced binding to GRB2. Reduced endocytosis by 50%. Reduced endocytosis by 77%; when associated with N-1203.
Functional Annotations	UP_SEQ_FEATURE	REGION:42 X 20 AA approximate tandem repeats of P-A-P-G-S-T-A-P-P-A-H-G-V-T-S-A-P-D-T-R
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with P53
Functional Annotations	UP_SEQ_FEATURE	REGION:Required for interaction with GSK3B
Functional Annotations	UP_SEQ_FEATURE	REGION:Required for interaction with beta- and gamma-catenins
Functional Annotations	UP_SEQ_FEATURE	REPEAT:10
Functional Annotations	UP_SEQ_FEATURE	REPEAT:11
Functional Annotations	UP_SEQ_FEATURE	REPEAT:12
Functional Annotations	UP_SEQ_FEATURE	REPEAT:13
Functional Annotations	UP_SEQ_FEATURE	REPEAT:14
Functional	UP_SEQ_FEATURE	REPEAT:15

Main Category	Category	Terms
Annotations		
Functional Annotations	UP_SEQ_FEATURE	REPEAT:16
Functional Annotations	UP_SEQ_FEATURE	REPEAT:17
Functional Annotations	UP_SEQ_FEATURE	REPEAT:18
Functional Annotations	UP_SEQ_FEATURE	REPEAT:19
Functional Annotations	UP_SEQ_FEATURE	REPEAT:1; approximate
Functional Annotations	UP_SEQ_FEATURE	REPEAT:20
Functional Annotations	UP_SEQ_FEATURE	REPEAT:21
Functional Annotations	UP_SEQ_FEATURE	REPEAT:22
Functional Annotations	UP_SEQ_FEATURE	REPEAT:23
Functional Annotations	UP_SEQ_FEATURE	REPEAT:24
Functional Annotations	UP_SEQ_FEATURE	REPEAT:25
Functional Annotations	UP_SEQ_FEATURE	REPEAT:26
Functional Annotations	UP_SEQ_FEATURE	REPEAT:27
Functional Annotations	UP_SEQ_FEATURE	REPEAT:28
Functional Annotations	UP_SEQ_FEATURE	REPEAT:29
Functional Annotations	UP_SEQ_FEATURE	REPEAT:2; approximate
Functional Annotations	UP_SEQ_FEATURE	REPEAT:3
Functional Annotations	UP_SEQ_FEATURE	REPEAT:30
Functional Annotations	UP_SEQ_FEATURE	REPEAT:31
Functional Annotations	UP_SEQ_FEATURE	REPEAT:32
Functional	UP_SEQ_FEATURE	REPEAT:33

Main Category	Category	Terms
Annotations		
Functional Annotations	UP_SEQ_FEATURE	REPEAT:34
Functional Annotations	UP_SEQ_FEATURE	REPEAT:35
Functional Annotations	UP_SEQ_FEATURE	REPEAT:36
Functional Annotations	UP_SEQ_FEATURE	REPEAT:37
Functional Annotations	UP_SEQ_FEATURE	REPEAT:38
Functional Annotations	UP_SEQ_FEATURE	REPEAT:39
Functional Annotations	UP_SEQ_FEATURE	REPEAT:4
Functional Annotations	UP_SEQ_FEATURE	REPEAT:40
Functional Annotations	UP_SEQ_FEATURE	REPEAT:41
Functional Annotations	UP_SEQ_FEATURE	REPEAT:42
Functional Annotations	UP_SEQ_FEATURE	REPEAT:43
Functional Annotations	UP_SEQ_FEATURE	REPEAT:44
Functional Annotations	UP_SEQ_FEATURE	REPEAT:45
Functional Annotations	UP_SEQ_FEATURE	REPEAT:46; approximate
Functional Annotations	UP_SEQ_FEATURE	REPEAT:47; approximate
Functional Annotations	UP_SEQ_FEATURE	REPEAT:48; approximate
Functional Annotations	UP_SEQ_FEATURE	REPEAT:5
Functional Annotations	UP_SEQ_FEATURE	REPEAT:6
Functional Annotations	UP_SEQ_FEATURE	REPEAT:7
Functional Annotations	UP_SEQ_FEATURE	REPEAT:8
Functional	UP_SEQ_FEATURE	REPEAT:9

Main Category	Category	Terms
Annotations		
Functional Annotations	UP_SEQ_FEATURE	SITE:Cleavage; by autolysis
Functional Annotations	UP_SEQ_FEATURE	TOPO_DOM:Cytoplasmic
Functional Annotations	UP_SEQ_FEATURE	TOPO_DOM:Extracellular
Functional Annotations	UP_SEQ_FEATURE	TRANSMEM:Helical
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of transcription by competitive promoter binding
Gene Ontology	GOTERM_BP_DIRECT	DNA damage response, signal transduction by p53 class mediator
Gene Ontology	GOTERM_BP_DIRECT	mitotic G1 DNA damage checkpoint signaling
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cell adhesion mediated by integrin
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of intrinsic apoptotic signaling pathway in response to DNA damage by p53 class mediator
Gene Ontology	GOTERM_CC_DIRECT	chromatin
Gene Ontology	GOTERM_CC_DIRECT	extracellular region
Gene Ontology	GOTERM_CC_DIRECT	extracellular space
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	Golgi lumen
Gene Ontology	GOTERM_CC_DIRECT	plasma membrane
Gene Ontology	GOTERM_CC_DIRECT	membrane
Gene Ontology	GOTERM_CC_DIRECT	apical plasma membrane
Gene Ontology	GOTERM_CC_DIRECT	vesicle

Main Category	Category	Terms
Gene Ontology	GOTERM_CC_DIRECT	extracellular exosome
Gene Ontology	GOTERM_MF_DIRECT	RNA polymerase II cis-regulatory region sequence-specific DNA binding
Gene Ontology	GOTERM_MF_DIRECT	p53 binding
Gene Ontology	GOTERM_MF_DIRECT	transcription coregulator activity
Gene Ontology	GOTERM_MF_DIRECT	protein binding
General Annotations	CHROMOSOME	1
General Annotations	CYTOGENETIC_LOCATION	1q22
General Annotations	ENTREZ_GENE_SUMMARY	This gene encodes a membrane-bound protein that is a member of the mucin family. Mucins are O-glycosylated proteins that play an essential role in forming protective mucous barriers on epithelial surfaces. These proteins also play a role in intracellular signaling. This protein is expressed on the apical surface of epithelial cells that line the mucosal surfaces of many different tissues including lung, breast stomach and pancreas. This protein is proteolytically cleaved into alpha and beta subunits that form a heterodimeric complex. The N-terminal alpha subunit functions in cell-adhesion and the C-terminal beta subunit is involved in cell signaling. Overexpression, aberrant intracellular localization, and changes in glycosylation of this protein have been associated with carcinomas. This gene is known to contain a highly polymorphic variable number tandem repeats (VNTR) domain. Alternate splicing results in multiple transcript variants.[provided by RefSeq, Feb 2011]
General Annotations	UP_COMMENT_ALTERNATIVE_PRODUCTS	Event=Alternative splicing; Named isoforms=17; Comment=Additional isoforms seem to exist.; Name=1; Synonyms=A; IsoId=P15941-1; Sequence=Displayed; Name=2; Synonyms=B; IsoId=P15941-2; Sequence=VSP_003280; Name=3; Synonyms=C; IsoId=P15941-3; Sequence=VSP_003281; Name=4; Synonyms=D; IsoId=P15941-4; Sequence=VSP_003282; Name=5; Synonyms=SEC; IsoId=P15941-5; Sequence=VSP_003288, VSP_003289; Name=6; Synonyms=X; IsoId=P15941-6; Sequence=VSP_003283, VSP_003284; Name=Y; Synonyms=MUC1/Y; IsoId=P15941-7; Sequence=VSP_003285; Name=8; Synonyms=Z; IsoId=P15941-8; Sequence=VSP_003286; Name=9; Synonyms=S; IsoId=P15941-9; Sequence=VSP_003286, VSP_003287; Name=F; IsoId=P15941-10; Sequence=VSP_035046, VSP_035047; Name=Y-LSP; IsoId=P15941-11; Sequence=VSP_003280, VSP_003285; Name=S2; IsoId=P15941-12; Sequence=VSP_003280,

Main Category	Category	Terms
		VSP_003285, VSP_003287; Name=M6; IsoId=P15941-13; Sequence=VSP_003286, VSP_046962, VSP_046963; Name=ZD; Synonyms=J19; IsoId=P15941-14; Sequence=VSP_047575, VSP_047576; Name=T10; IsoId=P15941-15; Sequence=VSP_003280, VSP_047873; Name=E2; IsoId=P15941-16; Sequence=VSP_003280, VSP_047872; Name=J13; IsoId=P15941-17; Sequence=VSP_003280, VSP_003285, VSP_047874;
General Annotations	UP_COMMENT_DEVELOPMENTAL_STAGE	During fetal development, expressed at low levels in the colonic epithelium from 13 weeks of gestation. {ECO:0000269|PubMed:9536947}.
General Annotations	UP_COMMENT_DISEASE	Note=MUC1/CA 15-3 is used as a serological clinical marker of breast cancer to monitor response to breast cancer treatment and disease recurrence (PubMed:20816948). Decreased levels over time may be indicative of a positive response to treatment. Conversely, increased levels may indicate disease progression. At an early stage disease, only 21% of patients exhibit high MUC1/CA 15-3 levels, that is why CA 15-3 is not a useful screening test. Most antibodies target the highly immunodominant core peptide domain of 20 amino acid (APDTRPAPGSTAPPAHGVTS) tandem repeats. Some antibodies recognize glycosylated epitopes. {ECO:0000269|PubMed:20816948}.
General Annotations	UP_COMMENT_DISEASE	Tubulointerstitial kidney disease, autosomal dominant 2 (ADTKD2) [MIM:174000]: A form of autosomal dominant tubulointerstitial kidney disease, a genetically heterogeneous disorder characterized by slowly progressive loss of kidney function, bland urinary sediment, hyperuricemia, absent or mildly increased albuminuria, lack of severe hypertension during the early stages, and normal or small kidneys on ultrasound. Renal histology shows variable abnormalities including interstitial fibrosis with tubular atrophy, microcystic dilatation of the tubules, thickening of tubular basement membranes, medullary cysts, and secondary glomerulosclerotic or glomerulocystic changes with abnormal glomerular tufting. There is significant variability, as well as incomplete penetrance. {ECO:0000269|PubMed:23396133}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_FUNCTION	The alpha subunit has cell adhesive properties. Can act both as an adhesion and an anti-adhesion protein. May provide a protective layer on epithelial cells against bacterial and enzyme attack.
General Annotations	UP_COMMENT_FUNCTION	The beta subunit contains a C-terminal domain which is involved in cell signaling, through phosphorylations and protein- protein interactions. Modulates signaling in ERK, SRC and NF-kappa-B pathways. In activated T-cells, influences directly or indirectly the Ras/MAPK pathway. Promotes tumor progression. Regulates TP53-mediated transcription and determines cell fate in the genotoxic stress response. Binds, together with KLF4, the PE21 promoter element of TP53 and represses

Main Category	Category	Terms
		TP53 activity.
General Annotations	UP_COMMENT_INTERACTION	P15941; P00519: ABL1; NbExp=4; IntAct=EBI-2804728, EBI-375543; P15941; P00533: EGFR; NbExp=4; IntAct=EBI-2804728, EBI-297353; P15941; P08581: MET; NbExp=2; IntAct=EBI-2804728, EBI-1039152; P15941; P15941-7: MUC1; NbExp=2; IntAct=EBI-2804728, EBI-4396776; P15941-11; Q08AM2: ADAM33; NbExp=3; IntAct=EBI-17263240, EBI-10225815; P15941-11; O60242: ADGRB3; NbExp=3; IntAct=EBI-17263240, EBI-2682765; P15941-11; Q15848: ADIPOQ; NbExp=3; IntAct=EBI-17263240, EBI-10827839; P15941-11; Q86W74-2: ANKRD46; NbExp=3; IntAct=EBI-17263240, EBI-12109402; P15941-11; P02652: APOA2; NbExp=3; IntAct=EBI-17263240, EBI-1171525; P15941-11; P05067-2: APP; NbExp=3; IntAct=EBI-17263240, EBI-17264467; P15941-11; P29972: AQP1; NbExp=3; IntAct=EBI-17263240, EBI-745213; P15941-11; P41181: AQP2; NbExp=3; IntAct=EBI-17263240, EBI-12701138; P15941-11; Q92482: AQP3; NbExp=3; IntAct=EBI-17263240, EBI-2808854; P15941-11; Q9H2C2: ARV1; NbExp=3; IntAct=EBI-17263240, EBI-11724186; P15941-11; Q92843: BCL2L2; NbExp=3; IntAct=EBI-17263240, EBI-707714; P15941-11; Q6PL45-2: BRICD5; NbExp=3; IntAct=EBI-17263240, EBI-12244618; P15941-11; Q8WVV5: BTN2A2; NbExp=3; IntAct=EBI-17263240, EBI-8648738; P15941-11; P06681: C2; NbExp=3; IntAct=EBI-17263240, EBI-2835920; P15941-11; O14523: C2CD2L; NbExp=3; IntAct=EBI-17263240, EBI-12822627; P15941-11; Q06432: CACNG1; NbExp=3; IntAct=EBI-17263240, EBI-9686780; P15941-11; Q9P0B6: CCDC167; NbExp=3; IntAct=EBI-17263240, EBI-9083477; P15941-11; Q08722-3: CD47; NbExp=3; IntAct=EBI-17263240, EBI-17263290; P15941-11; P19397: CD53; NbExp=3; IntAct=EBI-17263240, EBI-6657396; P15941-11; P34810: CD68; NbExp=3; IntAct=EBI-17263240, EBI-2826276; P15941-11; Q8N6F1-2: CLDN19; NbExp=3; IntAct=EBI-17263240, EBI-12256978; P15941-11; P56747: CLDN6; NbExp=3; IntAct=EBI-17263240, EBI-12955011; P15941-11; Q8NHS1: CLDND2; NbExp=3; IntAct=EBI-17263240, EBI-11959453; P15941-11; Q96FZ5: CMTM7; NbExp=3; IntAct=EBI-17263240, EBI-2807956; P15941-11; Q4VAQ0: COL8A2; NbExp=3; IntAct=EBI-17263240, EBI-10241815; P15941-11; Q8N6G5: CSGALNACT2; NbExp=3; IntAct=EBI-17263240, EBI-10267100; P15941-11; Q07325: CXCL9; NbExp=3; IntAct=EBI-17263240, EBI-3911467; P15941-11; O43169: CYB5B; NbExp=3; IntAct=EBI-17263240, EBI-1058710; P15941-11; P78329: CYP4F2; NbExp=3; IntAct=EBI-17263240, EBI-1752413; P15941-11; P56851: EDDM3B; NbExp=3; IntAct=EBI-17263240, EBI-10215665; P15941-11; Q9BV81: EMC6; NbExp=3; IntAct=EBI-17263240, EBI-2820492; P15941-11; P54852: EMP3; NbExp=3; IntAct=EBI-17263240, EBI-3907816; P15941-11; O75355-2: ENTPD3; NbExp=3; IntAct=EBI-17263240, EBI-12279764; P15941-11; Q9UKR5: ERG28; NbExp=3; IntAct=EBI-17263240, EBI-711490; P15941-11; P01350: GAST; NbExp=3;

Main Category	Category	Terms
		IntAct=EBI-17263240, EBI-3436637; P15941-11; P39905-3: GDNF; NbExp=3; IntAct=EBI-17263240, EBI-12702062; P15941-11; Q9Y3E0: GOLT1B; NbExp=3; IntAct=EBI-17263240, EBI-4402607; P15941-11; Q9NPR9: GPR108; NbExp=3; IntAct=EBI-17263240, EBI-11343451; P15941-11; Q9HCP6: HHATL; NbExp=3; IntAct=EBI-17263240, EBI-5916693; P15941-11; O60725: ICMT; NbExp=3; IntAct=EBI-17263240, EBI-11721771; P15941-11; Q9Y5U4: INSIG2; NbExp=3; IntAct=EBI-17263240, EBI-8503746; P15941-11; P11215: ITGAM; NbExp=3; IntAct=EBI-17263240, EBI-2568251; P15941-11; Q969L2: MAL2; NbExp=3; IntAct=EBI-17263240, EBI-944295; P15941-11; Q13021: MALL; NbExp=3; IntAct=EBI-17263240, EBI-750078; P15941-11; Q9P0N8: MARCHF2; NbExp=3; IntAct=EBI-17263240, EBI-10317612; P15941-11; Q6N075: MFSD5; NbExp=3; IntAct=EBI-17263240, EBI-3920969; P15941-11; P30301: MIP; NbExp=3; IntAct=EBI-17263240, EBI-8449636; P15941-11; Q96S97: MYADM; NbExp=3; IntAct=EBI-17263240, EBI-13301517; P15941-11; O95167: NDUFA3; NbExp=3; IntAct=EBI-17263240, EBI-1246131; P15941-11; Q99519: NEU1; NbExp=3; IntAct=EBI-17263240, EBI-721517; P15941-11; Q92982: NINJ1; NbExp=3; IntAct=EBI-17263240, EBI-2802124; P15941-11; Q9NZG7: NINJ2; NbExp=3; IntAct=EBI-17263240, EBI-10317425; P15941-11; Q16617: NKG7; NbExp=3; IntAct=EBI-17263240, EBI-3919611; P15941-11; Q8N912: NRAC; NbExp=3; IntAct=EBI-17263240, EBI-12051377; P15941-11; Q8NH19: OR10AG1; NbExp=3; IntAct=EBI-17263240, EBI-13339917; P15941-11; Q6TCH4: PAQR6; NbExp=3; IntAct=EBI-17263240, EBI-17265310; P15941-11; P26678: PLN; NbExp=3; IntAct=EBI-17263240, EBI-692836; P15941-11; P60201-2: PLP1; NbExp=3; IntAct=EBI-17263240, EBI-12188331; P15941-11; Q8IY26: PLPP6; NbExp=3; IntAct=EBI-17263240, EBI-11721828; P15941-11; P54315: PNLIPRP1; NbExp=3; IntAct=EBI-17263240, EBI-8652812; P15941-11; Q59EV6: PPGB; NbExp=3; IntAct=EBI-17263240, EBI-14210385; P15941-11; P30405: PPIF; NbExp=3; IntAct=EBI-17263240, EBI-5544229; P15941-11; Q96AA3: RFT1; NbExp=3; IntAct=EBI-17263240, EBI-6269616; P15941-11; Q02161-2: RHD; NbExp=3; IntAct=EBI-17263240, EBI-17249212; P15941-11; Q8TAC9: SCAMP5; NbExp=3; IntAct=EBI-17263240, EBI-2695784; P15941-11; Q9Y6D0: SELENOK; NbExp=3; IntAct=EBI-17263240, EBI-9679163; P15941-11; Q8N6R1: SERP2; NbExp=3; IntAct=EBI-17263240, EBI-749270; P15941-11; P11686: SFTPC; NbExp=3; IntAct=EBI-17263240, EBI-10197617; P15941-11; Q8IWU4: SLC30A8; NbExp=3; IntAct=EBI-17263240, EBI-10262251; P15941-11; Q969S0: SLC35B4; NbExp=3; IntAct=EBI-17263240, EBI-10281213; P15941-11; Q6ICL7: SLC35E4; NbExp=3; IntAct=EBI-17263240, EBI-12867720; P15941-11; Q9NVC3: SLC38A7; NbExp=3; IntAct=EBI-17263240, EBI-10314552; P15941-11; Q9NRQ5: SMCO4; NbExp=3; IntAct=EBI-17263240, EBI-8640191; P15941-11; B2RUZ4: SMIM1; NbExp=3; IntAct=EBI-17263240, EBI-12188413; P15941-11; Q9NZ01: TECR; NbExp=3;

Main Category	Category	Terms
		IntAct=EBI-17263240, EBI-2877718; P15941-11; P07204: THBD; NbExp=3; IntAct=EBI-17263240, EBI-941422; P15941-11; Q9BZW4: TM6SF2; NbExp=3; IntAct=EBI-17263240, EBI-13082040; P15941-11; P17152: TMEM11; NbExp=3; IntAct=EBI-17263240, EBI-723946; P15941-11; A0PK00: TMEM120B; NbExp=3; IntAct=EBI-17263240, EBI-10171534; P15941-11; Q9BTD3: TMEM121; NbExp=3; IntAct=EBI-17263240, EBI-12155101; P15941-11; Q5BJH2-2: TMEM128; NbExp=3; IntAct=EBI-17263240, EBI-10694905; P15941-11; Q9BVK8: TMEM147; NbExp=3; IntAct=EBI-17263240, EBI-348587; P15941-11; Q9Y6G1: TMEM14A; NbExp=3; IntAct=EBI-17263240, EBI-2800360; P15941-11; Q9P0S9: TMEM14C; NbExp=3; IntAct=EBI-17263240, EBI-2339195; P15941-11; Q14656: TMEM187; NbExp=3; IntAct=EBI-17263240, EBI-13046724; P15941-11; Q8NBD8: TMEM229B; NbExp=3; IntAct=EBI-17263240, EBI-12195227; P15941-11; Q9BU79: TMEM243; NbExp=3; IntAct=EBI-17263240, EBI-12887458; P15941-11; Q8N2M4: TMEM86A; NbExp=3; IntAct=EBI-17263240, EBI-12015604; P15941-11; Q8N661: TMEM86B; NbExp=3; IntAct=EBI-17263240, EBI-2548832; P15941-11; Q5BJF2: TMEM97; NbExp=3; IntAct=EBI-17263240, EBI-12111910; P15941-11; Q9Y2Y6: TMEM98; NbExp=3; IntAct=EBI-17263240, EBI-7333781; P15941-11; O14763: TNFRSF10B; NbExp=3; IntAct=EBI-17263240, EBI-518882; P15941-11; Q8N609: TRAM1L1; NbExp=3; IntAct=EBI-17263240, EBI-11996766; P15941-11; Q5BVD1: TTMP; NbExp=3; IntAct=EBI-17263240, EBI-10243654; P15941-11; Q53HI1: UNC50; NbExp=3; IntAct=EBI-17263240, EBI-7601760; P15941-11; O95183: VAMP5; NbExp=3; IntAct=EBI-17263240, EBI-10191195; P15941-11; Q9BQB6: VKORC1; NbExp=3; IntAct=EBI-17263240, EBI-6256462; P15941-11; Q8IVQ6: ZDHHC21; NbExp=3; IntAct=EBI-17263240, EBI-2849773; P15941-12; P00519: ABL1; NbExp=4; IntAct=EBI-34603716, EBI-375543; PRO_0000317447; P17676: CEBPB; NbExp=8; IntAct=EBI-10053698, EBI-969696;
General Annotations	UP_COMMENT_MISCELLANEOUS	The name KL-6 was originally that of a murine monoclonal antibody reacting with pulmonary adenocarcinoma cell lines and pulmonary epithelial cells. This antibody recognizes a sialylated carbohydrate chain on MUC1.
General Annotations	UP_COMMENT_MISCELLANEOUS	[Isoform Y-LSP]: Lacks the mucin repeats. {ECO:0000305}.
General Annotations	UP_COMMENT_MISCELLANEOUS	[Isoform ZD]: Lacks the mucin repeats. Exists as a disulfide-linked oligomer. {ECO:0000305}.
General Annotations	UP_COMMENT_POLYMORPHISM	The number of repeats is highly polymorphic. It varies from 21 to 125 in the northern European population. The most frequent alleles contains 41 and 85 repeats. The tandemly repeated icosapeptide underlies polymorphism at three positions: PAPGSTAP[PAQT]AHGVTSAP[DT/ES]R, DT -> ES and the single replacements P -> A, P -> Q and P-> T. The most

Main Category	Category	Terms
		frequent replacement DT -> ES occurs in up to 50% of the repeats.
General Annotations	UP_COMMENT_PTM	Dual palmitoylation on cysteine residues in the CQC motif is required for recycling from endosomes back to the plasma membrane. {ECO:0000269|PubMed:16507569}.
General Annotations	UP_COMMENT_PTM	Highly glycosylated (N- and O-linked carbohydrates and sialic acid). O-glycosylated to a varying degree on serine and threonine residues within each tandem repeat, ranging from mono- to penta- glycosylation. The average density ranges from about 50% in human milk to over 90% in T47D breast cancer cells. Further sialylation occurs during recycling. Membrane-shed glycoproteins from kidney and breast cancer cells have preferentially sialyated core 1 structures, while secreted forms from the same tissues display mainly core 2 structures. The O-glycosylated content is overlapping in both these tissues with terminal fucose and galactose, 2- and 3-linked galactose, 3- and 3,6- linked GalNAc-ol and 4-linked GlcNAc predominating. Differentially O-glycosylated in breast carcinomas with 3,4-linked GlcNAc. N- glycosylation consists of high-mannose, acidic complex-type and hybrid glycans in the secreted form MUC1/SEC, and neutral complex-type in the transmembrane form, MUC1/TM. {ECO:0000269|PubMed:10373415, ECO:0000269|PubMed:7744025, ECO:0000269|PubMed:9312074, ECO:0000269|PubMed:9597769}.
General Annotations	UP_COMMENT_PTM	Phosphorylated on tyrosines and serine residues in the C-terminal. Phosphorylation on tyrosines in the C-terminal increases the nuclear location of MUC1 and beta-catenin. Phosphorylation by PKC delta induces binding of MUC1 to beta-catenin/CTNNB1 and thus decreases the formation of the beta-catenin/E-cadherin complex. Src-mediated phosphorylation inhibits interaction with GSK3B. Src- and EGFR-mediated phosphorylation on Tyr-1229 increases binding to beta-catenin/CTNNB1. GSK3B-mediated phosphorylation on Ser-1227 decreases this interaction but restores the formation of the beta-cadherin/E-cadherin complex. On T-cell receptor activation, phosphorylated by LCK. PDGFR-mediated phosphorylation increases nuclear colocalization of MUC1CT and CTNNB1. {ECO:0000269|PubMed:11152665, ECO:0000269|PubMed:11483589, ECO:0000269|PubMed:11877440, ECO:0000269|PubMed:12750561, ECO:0000269|PubMed:14521915, ECO:0000269|PubMed:15513966, ECO:0000269|PubMed:16288032, ECO:0000269|PubMed:17545600, ECO:0000269|PubMed:7664271, ECO:0000269|PubMed:7988707, ECO:0000269|PubMed:9819408}.
General Annotations	UP_COMMENT_PTM	Proteolytic cleavage in the SEA domain occurs in the endoplasmic reticulum by an autoproteolytic mechanism and requires the full-length SEA domain as well as requiring a Ser, Thr or Cys residue at the P + 1 site. Cleavage at this site also occurs on isoform MUC1/X but not on isoform MUC1/Y. Ectodomain shedding is mediated by ADAM17. {ECO:0000269|

Main Category	Category	Terms
		PubMed:11341784, ECO:0000269|PubMed:12441351, ECO:0000269|PubMed:15987679}.
General Annotations	UP_COMMENT_PTM	The N-terminal sequence has been shown to begin at position 24 or 28. {ECO:0000269|PubMed:11341784}.
General Annotations	UP_COMMENT_SEQUENCE_CAUTION	Sequence=AAD14369.1; Type=Erroneous initiation; Note=Extended N-terminus.; Evidence={ECO:0000305}; Sequence=AAD14376.1; Type=Erroneous initiation; Note=Extended N-terminus.; Evidence={ECO:0000305};
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Apical cell membrane {ECO:0000256|ARBA:ARBA00004247}; Single-pass type I membrane protein {ECO:0000256|ARBA:ARBA00004247}. Cytoplasm {ECO:0000256|ARBA:ARBA00004496}. Nucleus {ECO:0000256|ARBA:ARBA00004123}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Apical cell membrane {ECO:0000269|PubMed:11118479, ECO:0000269|PubMed:12832415, ECO:0000269|PubMed:12939402, ECO:0000269|PubMed:15471854, ECO:0000269|PubMed:15972891, ECO:0000269|PubMed:16507569, ECO:0000269|PubMed:17524503, ECO:0000269|PubMed:17545600}; Single-pass type I membrane protein {ECO:0000269|PubMed:11118479, ECO:0000269|PubMed:12832415, ECO:0000269|PubMed:12939402, ECO:0000269|PubMed:15471854, ECO:0000269|PubMed:15972891, ECO:0000269|PubMed:16507569, ECO:0000269|PubMed:17524503, ECO:0000269|PubMed:17545600}. Note=Exclusively located in the apical domain of the plasma membrane of highly polarized epithelial cells. After endocytosis, internalized and recycled to the cell membrane. Located to microvilli and to the tips of long filopodial protusions.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Cell membrane {ECO:0000256|ARBA:ARBA00004236}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	[Isoform 5]: Secreted.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	[Isoform 9]: Secreted.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	[Isoform Y]: Secreted.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	[Mucin-1 subunit beta]: Cell membrane. Cytoplasm. Nucleus. Note=On EGF and PDGFRB stimulation, transported to the nucleus through interaction with CTNNB1, a process which is stimulated by phosphorylation. On HRG stimulation, colocalizes with JUP/gamma-catenin at the nucleus.
General Annotations	UP_COMMENT_SUBUNIT	The alpha subunit forms a tight, non-covalent heterodimeric complex with the proteolytically-released beta-subunit. Interaction, via the tandem repeat region, with domain 1 of ICAM1 is implicated in cell migration and metastases. Isoform 1 binds directly the SH2 domain of GRB2, and forms a MUC1/GRB2/SOS1 complex involved in RAS signaling. The cytoplasmic tail (MUC1CT) interacts with several proteins such as SRC,

Main Category	Category	Terms
		CTNNB1 and ERBs. Interaction with the SH2 domain of CSK decreases interaction with GSK3B. Interacts with CTNNB1/beta-catenin and JUP/gamma-catenin and promotes cell adhesion. Interaction with JUP/gamma-catenin is induced by heregulin. Binds PRKCD, ERBB2, ERBB3 and ERBB4. Heregulin (HRG) stimulates the interaction with ERBB2 and, to a much lesser extent, the interaction with ERBB3 and ERBB4. Interacts with P53 in response to DNA damage. Interacts with KLF4. Interacts with estrogen receptor alpha/ESR1, through its DNA-binding domain, and stimulates its transcription activity. Binds ADAM17. Isoform ZD forms disulfide-linked oligomers. {ECO:0000269|PubMed:11152665, ECO:0000269|PubMed:11173916, ECO:0000269|PubMed:11483589, ECO:0000269|PubMed:11877440, ECO:0000269|PubMed:12441351, ECO:0000269|PubMed:12750561, ECO:0000269|PubMed:12832415, ECO:0000269|PubMed:12939402, ECO:0000269|PubMed:14688481, ECO:0000269|PubMed:15471854, ECO:0000269|PubMed:15513966, ECO:0000269|PubMed:15710329, ECO:0000269|PubMed:16288032, ECO:0000269|PubMed:16427018, ECO:0000269|PubMed:16507569, ECO:0000269|PubMed:16983337, ECO:0000269|PubMed:17308127, ECO:0000269|PubMed:17524503, ECO:0000269|PubMed:7664271, ECO:0000269|PubMed:9139698, ECO:0000269|PubMed:9819408}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	Expressed on the apical surface of epithelial cells, especially of airway passages, breast and uterus. Also expressed in activated and unactivated T-cells. Overexpressed in epithelial tumors, such as breast or ovarian cancer and also in non-epithelial tumor cells. Isoform Y is expressed in tumor cells only. {ECO:0000269|PubMed:15513966, ECO:0000269|PubMed:9212228}.
Identifiers	ENTREZ_GENE_ID	4582
Identifiers	ENSEMBL_GENE_ID	ENSG00000185499
Identifiers	UNIPROT_ID	Q7Z540_HUMAN
Identifiers	UNIPROT_ID	Q7Z543_HUMAN
Identifiers	UNIPROT_ID	Q7Z546_HUMAN
Identifiers	UNIPROT_ID	Q7Z551_HUMAN
Identifiers	UNIPROT_ID	Q7Z541_HUMAN
Identifiers	UNIPROT_ID	Q7Z538_HUMAN
Identifiers	UNIPROT_ID	Q7Z539_HUMAN
Identifiers	UNIPROT_ID	Q7Z537_HUMAN
Identifiers	UNIPROT_ID	Q7Z545_HUMAN
Identifiers	UNIPROT_ID	Q7Z544_HUMAN
Identifiers	UNIPROT_ID	Q7Z549_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	Q7Z542_HUMAN
Identifiers	UNIPROT_ID	Q7Z536_HUMAN
Identifiers	UNIPROT_ID	Q14877_HUMAN
Identifiers	UNIPROT_ID	MUC1_HUMAN
Identifiers	UNIPROT_ID	B6ECB3_HUMAN
Identifiers	UNIPROT_ID	B6ECA2_HUMAN
Identifiers	UNIPROT_ID	B6ECB1_HUMAN
Identifiers	UNIPROT_ID	B6EC89_HUMAN
Identifiers	UNIPROT_ID	B6ECA4_HUMAN
Identifiers	UNIPROT_ID	B6ECB5_HUMAN
Identifiers	UNIPROT_ID	B6ECA3_HUMAN
Identifiers	UNIPROT_ID	B6EC96_HUMAN
Identifiers	UNIPROT_ID	B6ECB0_HUMAN
Identifiers	UNIPROT_ID	B6ECB2_HUMAN
Identifiers	UNIPROT_ID	B6ECA9_HUMAN
Identifiers	UNIPROT_ID	B4DWK6_HUMAN
Identifiers	UNIPROT_ID	B1AVQ7_HUMAN
Identifiers	UNIPROT_ID	A6ZID7_HUMAN
Identifiers	UNIPROT_ID	A6ZIF0_HUMAN
Identifiers	UNIPROT_ID	A6ZID8_HUMAN
Identifiers	UNIPROT_ID	A6ZIE7_HUMAN
Identifiers	UNIPROT_ID	A6ZID4_HUMAN
Identifiers	UNIPROT_ID	A6ZIE1_HUMAN
Identifiers	UNIPROT_ID	A6ZIE5_HUMAN
Identifiers	UNIPROT_ID	A6ZIE9_HUMAN
Identifiers	UNIPROT_ID	A6ZIE3_HUMAN
Identifiers	UNIPROT_ID	A6ZIE6_HUMAN
Identifiers	UNIPROT_ID	A6ZIF1_HUMAN
Identifiers	UNIPROT_ID	A6ZID5_HUMAN
Identifiers	UNIPROT_ID	A6ZIE8_HUMAN
Identifiers	UNIPROT_ID	A6ZID6_HUMAN
Identifiers	UNIPROT_ID	A6ZIE4_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	A6ZIE2_HUMAN
Identifiers	UNIPROT_ID	A5YRU3_HUMAN
Identifiers	UNIPROT_ID	A5YRU5_HUMAN
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Identifiers	UNIPROT_ID	A5YRU0_HUMAN
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Identifiers	UNIPROT_ID	A5YRV0_HUMAN
Identifiers	UNIPROT_ID	A5YRU9_HUMAN
Identifiers	UNIPROT_ID	A5YRV4_HUMAN
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Identifiers	UNIPROT_ID	A5YRV2_HUMAN
Identifiers	UNIPROT_ID	A0A384NPK6_HUMAN
Identifiers	UNIPROT_ID	A0A0C4DGW3_HUMAN
Identifiers	UNIPROT_ID	A0A0A0MRB3_HUMAN
Identifiers	UNIPROT_ID	A0A0A0MSH4_HUMAN
Identifiers	UNIPROT_ID	A0A087WVJ0_HUMAN
Identifiers	UNIPROT_ID	A0A087WTR1_HUMAN
Identifiers	UNIPROT_ID	A0A087X0L2_HUMAN
Identifiers	UNIPROT_ID	A0A087WWM2_HUMAN
Identifiers	UNIPROT_ID	A0A087X061_HUMAN
Identifiers	UNIPROT_ID	A0A087WZZ6_HUMAN
Identifiers	UNIPROT_ID	A0A087WZI8_HUMAN
Interactions	DRUGBANK	TG4010
Interactions	PUBCHEM	H-D-Arg-D-Arg-D-Arg-D-Arg-D-Arg-D-Arg-D-Arg-D-Arg-Arg-Cys-Gln-Cys-Arg-Arg-Lys-Asn-Tyr-Gly-Gln-Leu-Asp-Ile-Phe-Pro-OH
Literature	GENERIF_SUMMARY	MUC1 immunoreactivity correlates with malignant transformation in the colorectum
Literature	GENERIF_SUMMARY	marker for discriminating hepatocellular carcinoma from other carcinomas

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Tyrosine kinase c-Src constitutes a bridge between cystic fibrosis transmembrane regulator channel failure and MUC1 overexpression in cystic fibrosis
Literature	GENERIF_SUMMARY	The present results demonstrate that MUC1 associates with protein kinase Cdelta (PKCdelta).
Literature	GENERIF_SUMMARY	expression may be very useful predictor of surgical outcome in mass-forming intrahepatic cholangiocarcinoma
Literature	GENERIF_SUMMARY	2 new DHS are in the MUC1 promoter at -750 bp and -250 bp from the transcriptional start site. They are seen in human cell lines and a transgene. The -250 DHS is undetectable in cell lines from colon adenoma to carcinoma with reduced MUC1 expression.
Literature	GENERIF_SUMMARY	Endogenous MUC1 from T47D and MCF-7 cell supernatants revealed almost identical O-glycosylation profiles compared with the respective recombinant probes, indicating that the fusion proteins reflected the authentic O-glycan profiles of the cells
Literature	GENERIF_SUMMARY	Formation of MUC1 metabolic complex is conserved in tumor-derived and normal epithelial cells
Literature	GENERIF_SUMMARY	biosynthesis in breast cancer cell is altered by factors affecting cell proliferation
Literature	GENERIF_SUMMARY	Non-glycosylated tandem repeats of MUC1 facilitate attachment of breast tumor cells to normal human lung tissue and immobilized extracellular matrix proteins (ECM) in vitro and have a potential role in metastasis.
Literature	GENERIF_SUMMARY	Altered expression and allelic association of the hypervariable membrane mucin MUC1 in Helicobacter pylori gastritis.
Literature	GENERIF_SUMMARY	MUC1 is regulated by ZEB1 in epithelial cells
Literature	GENERIF_SUMMARY	Constitutive and inducible expression of MUC1 in human T cells
Literature	GENERIF_SUMMARY	CD227 is expressed on human and murine activated dendritic cells
Literature	GENERIF_SUMMARY	shedding is mediated by tumor necrosis factor-alpha converting enzyme/ADAM 17.
Literature	GENERIF_SUMMARY	The invasive behavior of breast cell lines may depend on different expression patterns of the MUCI gene determined by a genetic polymorphism.
Literature	GENERIF_SUMMARY	its cDNA used to vaccinate mice to suppress the development of lung metastases
Literature	GENERIF_SUMMARY	MUC1/beta-catenin interaction occurs in both primary and metastatic tumors, but is dramatically increased in metastatic lesions. Addition of MUC1-cytoplasmic domain peptides to breast cancer cell lines increases their invasive capability.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	MUC5AC, MUC1 and heparan sulfate proteoglycans cooperated in the formation of a biological inhibitory complex towards the function of E-cadherin in colonic neoplasms
Literature	GENERIF_SUMMARY	MUC1 may contribute to the anti-adhesive character of the tubal surface, inhibiting ectopic implantation.
Literature	GENERIF_SUMMARY	FUT1 catalyses the addition of alpha-1,2-fucose to MUC1 and MUC5AC apomucins
Literature	GENERIF_SUMMARY	results confirm that there is downregulation of MUC1 mucin expression in cancer culture cells treated with selective estrogen receptor modulators
Literature	GENERIF_SUMMARY	MUC1 expression was inhibited by NGF treatment in breast cancer tumor cell lines, suggesting that its expression can be regulated by signals resulting from the homodimerization of Erb-B2.
Literature	GENERIF_SUMMARY	It was also observed that the mucins from colon carcinoma patients had MUC1-type mucins that carried both sialyl-Lewis a and x epitopes and CD43-type sialyl-Lewis a mucins with only low levels of sialyl-Lewis x epitopes.
Literature	GENERIF_SUMMARY	Regulates intracellular oxidant levels and the apoptotic response to oxidative stress in cancer cells
Literature	GENERIF_SUMMARY	Two genes predisposing to dominant MCKD, MCKD1 and MCKD2, have been localized to chromosome 1q21 and 16p12. Review.
Literature	GENERIF_SUMMARY	A correlation has been found between EMA expression and prognosis in renal cell carcinoma patients who underwent nephrectomy.
Literature	GENERIF_SUMMARY	conformational features that contribute to acceptor substrate specificity
Literature	GENERIF_SUMMARY	findings demonstrate that MUC1 functions in cross talk between ErbB2 and Wnt pathways by acting as a shuttle for heregulin/neuregulin-1-induced nucleolar targeting of gamma-catenin
Literature	GENERIF_SUMMARY	These findings provide the first evidence that MUC1 induces cellular transformation.
Literature	GENERIF_SUMMARY	MUC1 and Met can be detected in the axillary fluids of patients with breast cancer; the expression of both tumor markers in the axillary drainage is strongly associated with unfavorable tumor features and can be used as a prognostic factor
Literature	GENERIF_SUMMARY	MUC1 splice variant expression is different in pure DCIS compared to DCIS with adjacent invasive cancer
Literature	GENERIF_SUMMARY	increased synthesis may be a key element in the host response to infection with oral pathogens
Literature	GENERIF_SUMMARY	All salivary gland tumors expressed MUC1.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	In breast cancer patients, MUC1 was detected both in tissue specimens as well as free in serum samples; furthermore, MUC1 can also circulate complexed with IgG and IgM antibodies; accurate measurements should include free and complexed forms.
Literature	GENERIF_SUMMARY	First simultaneous kinetic description of O-glycosylation events by human recombinant UDP-GalNAc
Literature	GENERIF_SUMMARY	MUC1 cytoplasmic domain coactivates Wnt target gene transcription and confers transformation.
Literature	GENERIF_SUMMARY	MUC1 synthesized in the airway cell line showed low levels of sialylation but carried a range of oligo-N-acetyllactosamine structures not seen in the colon carcinoma cell line.
Literature	GENERIF_SUMMARY	MUC1 integrates T cell receptor signaling with the beta-catenin pathway
Literature	GENERIF_SUMMARY	MUC1 attenuates the apoptotic response to DNA damage and that this oncoprotein confers resistance to genotoxic anticancer agents
Literature	GENERIF_SUMMARY	MUC1 mRNA expression was identified for twenty-six lymph nodes (15.7%), in eighteen patients (42%) who were diagnosed as having nodal micrometastasis.
Literature	GENERIF_SUMMARY	Mucin 1 and adenomatous polyposis coli co-immunoprecipitate in breast cancer cells.
Literature	GENERIF_SUMMARY	T-cell lines generated using the agonist epitope were able to lyse HLA-A2 human tumor cells expressing MUC-1.
Literature	GENERIF_SUMMARY	MUC1 and beta-catenin have a role in progression and invasiveness of colorectal carcinomas
Literature	GENERIF_SUMMARY	TNFalpha-stimulated MUC1 shedding occurs independently of increased de novo protein synthesis and TNFalpha-induced increase in MUC1 gene expression is mediated through the kappaB site in the MUC1 promoter
Literature	GENERIF_SUMMARY	VNTR polymorphism of the MUC1 gene was not associated with female infertility. Other regulatory molecules and signals may interact with the MUC1 gene variations, favouring endometrial receptivity and embryo attachment.
Literature	GENERIF_SUMMARY	Data show that MUC1 mucin expressed on the surface of cells can initiate a calcium-based oscillatory signal on contact with intercellular adhesion molecule (ICAM)-1-transfected cells.
Literature	GENERIF_SUMMARY	tumor-derived MUC-1 profoundly affects the cytokine repertoire of monocyte-derived DC and switch them into IL-10(high)IL-12(low) regulatory APCs with a limited capacity to trigger protective Th1 responses
Literature	GENERIF_SUMMARY	MUC1 regulates the FOXO3a signaling pathway in a

Main Category	Category	Terms
		survival response to oxidative stress.
Literature	GENERIF_SUMMARY	Grb2 plays a significant role in the endocytosis of MUC1
Literature	GENERIF_SUMMARY	Cytokeratin 7 and epithelial membrane antigen are essentially negative in yolk sac tumors but are diffusely positive in clear cell carcinomas and endometriod adenocarcinomas making them useful markers for differentiating YSTs from both CCCs and EACs.
Literature	GENERIF_SUMMARY	overexpression of MUC1 in pancreatic cancer is a useful target
Literature	GENERIF_SUMMARY	polymorphism in the MUC1 tandem repeat influences the expression of tumor-associated carbohydrate antigens in gastric cancer cells and may therefore allow the identification of subgroups of patients that develop more aggressive tumors expressing T antigen
Literature	GENERIF_SUMMARY	analysis of a novel MUC1 protein isoform MUC1/ZD, which is generated by an alternative splicing event that both deletes the tandem-repeat array and leads to a C-terminal reading frameshift
Literature	GENERIF_SUMMARY	IL-7 up-regulates MUC1 (CD227) on CD4+, CD8+, CD25+, CD69+, naive CD45RA+, and memory CD45RO+ T cells. IL-7, but not IL-2 or IL-4, markedly induces MUC1 expression on CD3+ T cells.
Literature	GENERIF_SUMMARY	MUC1 regulates p53-responsive genes and thereby cell fate in the genotoxic stress response
Literature	GENERIF_SUMMARY	The Muc1 Mucin is a large membrane-tethered glycoprotein that shows differential expression in many adenocarcinomas, where it contributes to their invasive and metastatic properties.
Literature	GENERIF_SUMMARY	MUC1 expression is related to tumor progression and worse prognosis in mucoepidermoid carcinomas of the salivary glands.
Literature	GENERIF_SUMMARY	evidence for a strong Th1 Cells polarization and cross-priming of MUC1-specific CD8+ T cells
Literature	GENERIF_SUMMARY	The altered dendritic cell differentiation and antigen presentation function induced by soluble sialylated tumor-associated mucin may represent a mechanism by which epithelial tumors can escape immunosurveillance.
Literature	GENERIF_SUMMARY	Results suggest that the large MUC1 allele is associated with susceptibility to lung adenocarcinoma and poor prognosis.
Literature	GENERIF_SUMMARY	Progesterone-dependent regulation of MUC1 appears to be an important factor in determining endometrial receptivity
Literature	GENERIF_SUMMARY	The human mucin MUC1 is expressed both as a transmembrane heterodimeric protein complex that recycles via the trans-Golgi network (TGN) and as a secreted isoform.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The MUC1 SEA module is a self-cleaving domain.
Literature	GENERIF_SUMMARY	MUC1 attracts immature dendritic cells in vivo in cancer patients through chemotaxis and subverts their type 1 helper T cell immunostimulatory function important for tumor rejection
Literature	GENERIF_SUMMARY	glycopeptide-specific helper T-cell clones are a prerequisite for strong and long-lasting immune responses to MUC1-positive tumours.
Literature	GENERIF_SUMMARY	Expression and localization of MUC1 proteins in primary liver carcinomas may act as prognostic markers.
Literature	GENERIF_SUMMARY	HNE stimulation of mucin release by human airway epithelial cells involves intracellular activation of PKC, specifically the delta isoform.
Literature	GENERIF_SUMMARY	Lack of MUC1 polymorphism in breast cancer is significantly related to neoplasm aggressiveness, especially lymphatic vessel invasion and lymph node metastasis.
Literature	GENERIF_SUMMARY	MUC1 is delivered to mitochondria by a mechanism involving activation of the ErbB receptor-->c-Src pathway and transport by the molecular chaperone HSP70/HSP90 complex
Literature	GENERIF_SUMMARY	MUC1, MUC5B and MUC8, but not MUC2 or MUC5AC, are up-regulated in endometrial adenocarcinomas
Literature	GENERIF_SUMMARY	Eighty-six patients (43%) had elevated CA 125 levels and 35 (17.5%) had elevated CA 15.3 levels at diagnosis
Literature	GENERIF_SUMMARY	Analysis of the ability of human milk to inhibit the attachment of rNV VLPs (recombinant NV-like particles) to their carbohydrate ligands and to characterize potential inhibitors found in milk is presented.
Literature	GENERIF_SUMMARY	Immunity and tumor protection in mice transgenic for human MUC.1, a glycoprotein expressed in the majority of epithelial tumors in humans, were induced by vaccination with B lymphocytes genetically programmed to activate MUC.1-specific CD4 T cells.
Literature	GENERIF_SUMMARY	pro-proliferative and anti-apoptotic effects of KL-6 on lung fibroblasts; findings suggest a novel pathophysiological role for KL-6 in idiopathic pulmonary fibrosis, as it stimulates fibrotic processes in the alveolar space
Literature	GENERIF_SUMMARY	ICAM-1 can influence the migration of cells that express endogenous or transfected MUC1.
Literature	GENERIF_SUMMARY	MUC1 568 A to G polymorphism may be of interest for diagnostic purposes because our study delivered in vivo evidence that it contributes to interindividual variations in KL-6 levels.
Literature	GENERIF_SUMMARY	Decrease of mucin contained in saliva may be important in further evolution or progression of

Main Category	Category	Terms
		squamous cell carcinoma oral cavity
Literature	GENERIF_SUMMARY	SEA domain of the human MUC1 transmembrane mucin undergoes a novel type of autoproteolysis, which is catalyzed by conformational stress and the conserved serine hydroxyl
Literature	GENERIF_SUMMARY	using RT-PCR for the detection of c-Met or MUC1 mRNA may be a promising tool for the early detection of micro-metastatic circulating tumor cells in gastric carcinoma patients
Literature	GENERIF_SUMMARY	MUC1 binds directly to the ERalpha DNA binding domain and stabilizes ERalpha by blocking its ubiquitination and degradation.
Literature	GENERIF_SUMMARY	Overexpresssed in the progression and lymphatic metastasis of prostate cancer.
Literature	GENERIF_SUMMARY	Reversed apical pattern of MUC1 expression is associated with invasive micropapillary carcinoma of the breast
Literature	GENERIF_SUMMARY	MUC1/Muc1 suppresses pulmonary innate immunity and speculate its anti-inflammatory activity may play an important modulatory role during microbial infection
Literature	GENERIF_SUMMARY	A marked difference is seen between the N-glycans on membrane-bound and secreted forms of the MUC1 native molecule.
Literature	GENERIF_SUMMARY	Expression of the tumor-associated Ag mucin-1 (MUC1) was detected using Monoclonal Antibody defining different MUC1 glycoforms.
Literature	GENERIF_SUMMARY	CA 15-3 expression increases in patients with advancing metastatic breast cancer
Literature	GENERIF_SUMMARY	results support the hypothesis that MUC1 contributes significantly to growth and metastasis, and that down-regulation of MUC1 protein expression decreases the metastatic potential of pancreatic adenocarcinoma
Literature	GENERIF_SUMMARY	MUC1 carries selectin ligands throughout the secretory phase of the menstrual cycle, including the mid-secretory (receptive) phase
Literature	GENERIF_SUMMARY	Progesterone receptor isoforms A and B differentially regulate MUC1 expression in uterine epithelial cells.
Literature	GENERIF_SUMMARY	atypical MUC1 bearing the noncluster form of Tn antigen is implicated in aggressive growth of primary breast cancer cells, particularly in lymphatic metastasis
Literature	GENERIF_SUMMARY	MUC1 causes varying changes in genetic transcription, cell proliferation, apoptosis,and invasion in 2 different breast cancer cell lines
Literature	GENERIF_SUMMARY	mucin and MUC1 have roles in progression of hereditary nonpolyposis colorectal cancer and sporadic colorectal cancer

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	MUC1 sequesters c-Abl in the cytoplasm and thereby inhibits apoptosis in the response to genotoxic anticancer agents.
Literature	GENERIF_SUMMARY	Disruption of MUC1-C-terminal subunitmight represent a novel immunotherapeutic approach for the treatment of ovarian cancer(review).
Literature	GENERIF_SUMMARY	Serum measurement has the potential to serve as a noninvasive diagnostic test for the detection of bronchiolitis obliterans syndrome in lung transplantation.
Literature	GENERIF_SUMMARY	data indicate that MUC1 expression is a potent regulator of erbB1 receptor stability upon activation and may promote transformation through the inhibition of erbB1 degradation
Literature	GENERIF_SUMMARY	The study demonstrates that the GSTA region of the MUC1 tandem repeat contains a highly immunodominant epitope when presented with immature short O-glycans.
Literature	GENERIF_SUMMARY	Poorly differentiated tumors are inversely correlated with tumor and serum MUC1 detection.
Literature	GENERIF_SUMMARY	GATA3 is probably a mediator for the transcriptional upregulation of MUC1 expression in some breast cancers.
Literature	GENERIF_SUMMARY	Galectin-3 interaction with Thomsen-Friedenreich disaccharide on cancer-associated MUC1 causes increased cancer cell endothelial adhesion
Literature	GENERIF_SUMMARY	MUC1 attenuates HIF-1alpha activation in a survival response to hypoxic stress
Literature	GENERIF_SUMMARY	The present studies show that MUC1 associates with FGFR3.
Literature	GENERIF_SUMMARY	The overexpression and membrane delocalization of MUC1 is associated with a worse prognosis and a shorter survival in breast, colon, kidney, prostate or gastro-intestinal cancers. [REVIEW]
Literature	GENERIF_SUMMARY	To validate the occurrence of MGL-MUC1 interactions in situ, we studied the binding of MGL to MUC1 in primary colon carcinoma tissue. Isolation of MUC1 out of colon carcinoma tissue showed strong binding activity to MGL.
Literature	GENERIF_SUMMARY	MUC1 peptide immunogenicity depends on context of the epitope and not sequence alone
Literature	GENERIF_SUMMARY	Expressions of membrane (MUC1) and secreted (MUC5AC, MUC6) mucins are frequently modified in reactive gastropathy.
Literature	GENERIF_SUMMARY	Histiocytoid features of invasive lobular carcinoma is immunopositive for MUC1.
Literature	GENERIF_SUMMARY	Interaction of MUC1 with CTNNB modulates cyclin D1

Main Category	Category	Terms
		in H. pylori-induced stomach neoplasms.
Literature	GENERIF_SUMMARY	Upregulation of MUC1 is associated with disease progression to adenocarcinoma of the esophagus
Literature	GENERIF_SUMMARY	A novel classification of MUC1 expression pattern (HP, LP, and D) was correlated with tumor differentiation and postoperative survival in NSCLC, especially in lung adenocarcinoma.
Literature	GENERIF_SUMMARY	Short mucin 1 alleles are associated with low-virulece H pylori strains infection.
Literature	GENERIF_SUMMARY	Review discusses the role of MUC1 overexpression in the progression of prostate cancer, and targeting MUC1 to control micrometastases and hormone refractory disease.
Literature	GENERIF_SUMMARY	Results suggest that MUC1 inhibits cell proliferation through a beta-catenin/LEF-1/cyclin D1/c-Myc pathway.
Literature	GENERIF_SUMMARY	Circulating TARC/CCL17 and KL-6 are useful measurements for discriminating acute eosinophilic pneumonia from other causes of acute lung injury.
Literature	GENERIF_SUMMARY	TNF-alpha induces MUC1 gene transcription through a TNFR1 --> MEK1/2 --> ERK1 --> Sp1 pathway.
Literature	GENERIF_SUMMARY	Results suggest that there is no effect of the polymorphic MUC-1 sequence on the implantation failure. However, the data do not exclude MUC-1 relevance during embryo implantation.
Literature	GENERIF_SUMMARY	This study shows that MUC1, MUC4, and MUC16 are regulated differently by dexamethasone in human corneal epithelial cells.
Literature	GENERIF_SUMMARY	EMA was not expressed in the germinative cells of sebaceoma, but was expressed strongly in approximately 50% of mature sebocytes; expression of EMA was uncommon in basal cell carcinoma and was confined to keratotic or squamoid areas
Literature	GENERIF_SUMMARY	Up-regulation of MUC1 by inflammatory mediators such as neutrophil elastase and TNF-alpha suggests a crucial role for MUC1 in the control of excessive inflammation during airway bacterial infection.
Literature	GENERIF_SUMMARY	bacterium affects the soluble form of MUC 1 mucin, thus suggesting a likely role of this mucin in the course of H. pylori infection
Literature	GENERIF_SUMMARY	IL-10(-/-) mice, crossed to human MUC1-transgenic mice, develop MUC1(+) IBD characterized by an earlier age of onset, higher inflammation scores, and a much higher incidence and number of colon cancers
Literature	GENERIF_SUMMARY	In stepwise multivariate analysis MUC1 negativity was an independent predictor of both relapse-free survival and overall survival.
Literature	GENERIF_SUMMARY	MUC1, TF and galectin might have important roles in

Main Category	Category	Terms
		endometrial pathogenesis and malignant transformation
Literature	GENERIF_SUMMARY	Four patterns of MUC1 expression are recognized in DCIS that suggest a relationship to functional differentiation.
Literature	GENERIF_SUMMARY	Typical cystitis glandularis expresses MUC1.
Literature	GENERIF_SUMMARY	glycosylation variants of MUC1 were expressed in the fetal respiratory tracts as early as 13 weeks of gestation, and its expression persisted even after lung maturation
Literature	GENERIF_SUMMARY	69% of the thalassaemic patients had abnormal levels of Carbohydrate antigen 15.3(CA 15.3)
Literature	GENERIF_SUMMARY	both the MUC1-N and MUC1-C subunits evolved from secreted gel-forming mucins and that the MUC1-CD oncogenic function emerged by diversification after evolution from MUC5B
Literature	GENERIF_SUMMARY	MUC1 expression was significantly reduced in the fallopian tubes of women with ectopic pregnancy.
Literature	GENERIF_SUMMARY	MUC1 in serum of ovarian and breast cancer patients inhibits huHMFG-1 dependent cell-mediated cytotoxicity
Literature	GENERIF_SUMMARY	PIASy has a role in down-regulation of MUC1 expression
Literature	GENERIF_SUMMARY	MUC-1 is an independent prognostic marker for prostate cancer death.
Literature	GENERIF_SUMMARY	MUC1 may affect cancer cell migration by increasing E-cadherin/beta-catenin complex formation and restoring E-cadherin membrane localization
Literature	GENERIF_SUMMARY	Measurement of CA 15-3 serum values in conjunction with sHER2 and CA 15-3 can increase sensitivity in metastasis detection.
Literature	GENERIF_SUMMARY	These findings indicate that MUC1 and galectin-3 function as part of a miR-322-dependent regulatory loop.
Literature	GENERIF_SUMMARY	The results indicated that IL-4-treated LS174T cells are able to produce mucins with a higher degree of O-glycosylation than untreated counterparts.
Literature	GENERIF_SUMMARY	A high correlation between MUC1 and cyclin D1 implies MUC1 involvement in the Wnt cascade functioning in a large subset of human papillary thyroid carcinoma and papillary microcarcinoma
Literature	GENERIF_SUMMARY	We found MUC1 expressed in male germ cell line and within the ejaculated sperm, but its presence in mature sperm does not seem to be related to male fertility.
Literature	GENERIF_SUMMARY	The present study failed to demonstrate any association between serum levels of MMPs, VEGF and CA 15.3 (antigen product from MUC1 gene) and well-

Main Category	Category	Terms
		known clinicopathological characteristics of breast carcinoma
Literature	GENERIF_SUMMARY	These results support the hypothesis that the adhesive interactions between MUC1 and MAG are of biological significance in pancreatic cancer perineural invasion.
Literature	GENERIF_SUMMARY	In colorectal adenocarcinoma, MUC1 expression was negatively correlated to E-cad expression.
Literature	GENERIF_SUMMARY	If MUC1 is detected in regional lymph nodes from patients with esophageal cancer. MUC1 does not appear to be expressed in normal lymph nodes.
Literature	GENERIF_SUMMARY	MUC1 is important for physiological activation of IKKbeta and that overexpression of MUC1, as found in human cancers, confers sustained induction of the IKKbeta-NF-kappaB p65 pathway.
Literature	GENERIF_SUMMARY	PKC delta plays an important role in mucin secretion by airway epithelium via regulation of MARCKS phosphorylation.
Literature	GENERIF_SUMMARY	LL-37 can directly stimulate mucus synthesis through activation of MUC1 and MUC2 expression and MAP kinase pathway in human colonic cells.
Literature	GENERIF_SUMMARY	To clarify the localization of KL-6 and epithelial membrane antigen (EMA) in human lungs, immune reactions to antibodies to these factors were examined in detail at light and electron microscopic levels.
Literature	GENERIF_SUMMARY	MUC1 stabilizes Bcr-Abl and contributes to pathogenesis of chronic myelogenous leukemia cells by promoting self renewal and the MUC1 cytoplasmic domain.
Literature	GENERIF_SUMMARY	The PAM4-reactive MUC1 epitope was not detected in normal pancreas but was expressed in invasive pancreatic adenocarcinomas, including early stage 1 disease
Literature	GENERIF_SUMMARY	Bile acids induce MUC1 mucin overexpression in human esophageal adenocarcinoma cells and tissues by activating its transcription through a process involving phosphatidylinositol 3-kinase.
Literature	GENERIF_SUMMARY	CA 15-3 levels might have a role in the response to EGFR inhibitors in patients with bronchioloalveolar carcinoma
Literature	GENERIF_SUMMARY	Genetic polymorphisms in the glycosylation process of MUC1 may be novel risk factors for ovarian cancer.
Literature	GENERIF_SUMMARY	SEA domain autoproteolysis accelerated by conformational strain.
Literature	GENERIF_SUMMARY	SEA domain autoproteolysis accelerated by conformational strain
Literature	GENERIF_SUMMARY	MUC1 mucin variability conditions H pylori binding to gastric cells. The extent of bacterial adhesion depends

Main Category	Category	Terms
		on the size of the MUC1 VNTR domain.
Literature	GENERIF_SUMMARY	demonstrate for the first time that the near-ubiquitous MUC1 tumor Ag can be targeted using chimeric Ag receptors-grafted T cells
Literature	GENERIF_SUMMARY	Breast cancer cells express MUC1 and give rise to MUC1(+) tumors in vivo. MUC1 is hypoglycosylated and sialylated; characteristics of the tumor-specific form were expressed on cancer cells and recognized by tumor-specific T cells and antibodies.
Literature	GENERIF_SUMMARY	These findings suggest a novel MUC1-Src-CrkL-Rac1/Cdc42 signaling cascade following ICAM-1 ligation, through which MUC1 regulates cytoskeletal reorganization and directed cell motility during cell migration.
Literature	GENERIF_SUMMARY	analysis of the role of MUC1 in colorectal cancer [review editorial]
Literature	GENERIF_SUMMARY	Epigenetic changes of MUC1 may be of importance for diagnosis of carcinogenic risk and prediction of outcome for cancer patients.
Literature	GENERIF_SUMMARY	MUC1 expression in the human placenta is increased during placental development, and its overexpression suppresses trophoblast-like cell invasion in vitro.
Literature	GENERIF_SUMMARY	Evidence is presented that strongly supports a mechanism whereby dimerization of the extracellular domain of MUC1* activates the MAP kinase signaling cascade and stimulates cell growth.
Literature	GENERIF_SUMMARY	CA15.3 (soluble MUC1) levels were significantly higher in the serum of mesothelioma patients than in healthy controls but were not significantly different to levels in patients with benign asbestos-related disease.
Literature	GENERIF_SUMMARY	MUC-1 expression was found to be a prognostic factor for gastric carcinoma patients, albeit not independent of parameters of invasion.
Literature	GENERIF_SUMMARY	MUC1 was expressed more intensely in prostate intraepithelial neoplasia and malignant prostate glands than in benign glands.
Literature	GENERIF_SUMMARY	MUC1 activates the beta-catenin and NF-kappaB pathways in multiple myeloma cells and contributes to their growth and survival.
Literature	GENERIF_SUMMARY	Bladder cancer stem (initiating) cells might be among EMA(-) CD44v6(+) subset.
Literature	GENERIF_SUMMARY	Serum samples were investigated for carcinoembryonic antigen (CEA), CA125 and MUC1, alpha-foetoprotein, neuron-specific enolase and CA19.9.
Literature	GENERIF_SUMMARY	MUC1 mucin is present in several glycoforms on the maternal side of human placenta after term delivery
Literature	GENERIF_SUMMARY	Phosphorylation of MUC1 by Met modulates interaction

Main Category	Category	Terms
		with p53 and MMP1 expression.(
Literature	GENERIF_SUMMARY	Common variants in MUC1 and the surrounding region are not associated with risk or prognosis of prostate cancer in Swedish men.
Literature	GENERIF_SUMMARY	Retroviral overexpression of Mucin1 in cord blood CD34+ cells resulted in elevated stem cell and progenitor frequencies.
Literature	GENERIF_SUMMARY	MUC1 is of importance to the physiologic regulation of caspase-8 activity and that overexpression of MUC1, as found in human malignancies, could contribute to constitutive inhibition of death receptor signaling pathways
Literature	GENERIF_SUMMARY	Significant enhancement in the development of pancreatic intraepithelial preneoplastic lesions and progression to adenocarcinoma is observed in PDA.MUC1 transgenic mice
Literature	GENERIF_SUMMARY	These results indicated that MUC1 intra-cellular trafficking occurs through a regulated process that was stimulated by direct EGFR and MUC1 interaction, mediated by clathrin coated pits that were dynamin dependent and regulated by Rab5.
Literature	GENERIF_SUMMARY	immunotoxin binding to the MUC1 alpha/beta junction mediates cell killing
Literature	GENERIF_SUMMARY	Stimulating the MUC1* receptor with either the cognate antibody or its ligand NM23 enabled hESC growth in a feeder cell-free system and produced pluripotent colonies that resisted spontaneous differentiation
Literature	GENERIF_SUMMARY	These mucin genes were up-regulated after exposure to low pH in vitro (p
Literature	GENERIF_SUMMARY	Alternatively glycosylated MUC1 and MUC5AC could be of significant potential as effective tumor markers in gastric cancer diagnosis.
Literature	GENERIF_SUMMARY	the combined pattern of positive KL-6 expression and decreased membranous beta-catenin expression by colorectal carcinoma is a useful biomarker for distinguishing a subgroup of patients with a worse prognosis.
Literature	GENERIF_SUMMARY	Surface MUC1 on Dendritic cells does not influence intrinsic cell-motility, nor is it involved in cell-cell and cell-matrix dependent migration.
Literature	GENERIF_SUMMARY	Serum CA 15-3 was found to have a value in the early detection and monitoring of treatment of breast cancer in Ghana
Literature	GENERIF_SUMMARY	MUC1 tandem repeat glycopeptides are capable of activating both helper and cytotoxic T-cells
Literature	GENERIF_SUMMARY	MUC1 was present in urothelial cell carcinoma in situ (CIS) of the bladder without a polarized staining pattern and was expressed more strongly in cases with

Main Category	Category	Terms
		multifocal disease.
Literature	GENERIF_SUMMARY	MUC1 is overexpressed and hypoglycosylated in pediatric inflammatory bowel disease.
Literature	GENERIF_SUMMARY	Human eccrine sweat glands express CK7, CK8, CK14, CK18, CK19, CEA, EMA, Ki67, p63, EGF and EGFR. In skin, CEA can be used as a specific immunological marker of sweat glands.
Literature	GENERIF_SUMMARY	Significant differences were observed between mean CA 15-3 levels of control subjects grouped according to the MUC1 568 genotype (mean+/-SD)
Literature	GENERIF_SUMMARY	Binding of the MUC1 CT to CAML in human epithelial cells was confirmed by reciprocal coimmunoprecipitations, confocal microscopy, protein crosslinking, and coupled transcription/translation analyses.
Literature	GENERIF_SUMMARY	These data suggest that hypoxia enhances the expression of MUC1 through the transcriptional regulation by HIF-1alpha in a human lung epithelial cell line.
Literature	GENERIF_SUMMARY	MUC1 as a novel MAL2 partner (encoding a four-transmembrane protein of the MAL family)
Literature	GENERIF_SUMMARY	Macrophage-derived interleukin-6 up-regulates MUC1, but down-regulates MUC2 expression in colon cancer.
Literature	GENERIF_SUMMARY	evaluated IgM and IgG anti-MUC1 antibodies in 148 subjects with non-advanced adenomas, advanced adenomas, colorectal cancer, hyperplastic polyps, and normal controls; the prevalence of anti-MUC1 antibodies increase along the adenoma-carcinoma sequence
Literature	GENERIF_SUMMARY	In pancreatobiliary adenocarcinomas, MUC1 expression
Literature	GENERIF_SUMMARY	both foetal and adult tissues of known genotype, there is clear evidence for the role of single-nucleotide polymorphism in controlling alternative splicing of the 5' exon 2 region of both full-length transcripts and those lacking the tandem repeat domain.
Literature	GENERIF_SUMMARY	Invasive micropapillary carcinoma more commonly showed immunoreactivity for MUC1, CA125, and Her2Neu compared to invasive urothelial carcinoma with retraction artifact
Literature	GENERIF_SUMMARY	Serum levels of SP-D and KL-6 appear to be indicative of ""alveolitis"" in systemic sclerosis patients, and are significantly higher than in SSc patients without ""alveolitis.""""
Literature	GENERIF_SUMMARY	EMA is a better marker than XIAP or GLUT-1 for the diagnosis of malignant mesothelioma
Literature	GENERIF_SUMMARY	MUC1 regulates cholesterol and fatty acid metabolism, and activation of these pathways in ER(+) breast cancers predicts failure to tamoxifen treatment.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Report the value of CA 15-3 determination in the initial management of breast cancer patients.
Literature	GENERIF_SUMMARY	A 35-gene MUC1-induced tumorigenesis signature predicts significant decreases in both disease-free and overall survival in patients with breast (n=295) and lung (n=442) cancers.
Literature	GENERIF_SUMMARY	The utility of the tumour markers carcinoembryonic antigen, cancer antigen 125, carbohydrate antigen 19-9, carbohydrate antigen 15-3, alpha-fetoprotein and human chorionic gonadotropin for the diagnosis solitary pulmonary nodules, was investigated.
Literature	GENERIF_SUMMARY	Results show that circulating neoplastic cells keep a significant prognosis value at long term on high risk breast cancer patients, and that a double detection of MUC1 and CK19 does not improve the prognostic value of circulating neoplastic cells.
Literature	GENERIF_SUMMARY	repetitive units of Lewis X, within the mucin domain, play an important role in inhibiting transmission of HIV-1 from mother to child.
Literature	GENERIF_SUMMARY	MUC1 expression is correlated with the histological degree of malignancy in prostate cancer
Literature	GENERIF_SUMMARY	The anti-apoptotic effect of MUC1* overexpression may be the primary mechanism by which some breast cancer cells become resistant to chemotherapeutic agents. MUC1 associates with HER2 and plays a role in cross talk between the ErbB2 and Wnt pathways.
Literature	GENERIF_SUMMARY	PankoMab revealed strong correlation to hormone receptor expression in ductal carcinoma of the breast.
Literature	GENERIF_SUMMARY	Fibrinogen binding to ICAM-1 promotes mucin production by decreasing TGF-alpha-induced EGFR and ERK1/2 activation.
Literature	GENERIF_SUMMARY	the percentage of lung carcinoma patients remaining unclassifiable by TTF-1/TP63 was twice that of the five-antibody (TRIM29, CEACAM5, SLC7A5, MUC1, and CK5/6) test
Literature	GENERIF_SUMMARY	Serum KL-6 and SP-D appear to be elevated in a mutually exclusive manner in adult T-cell leukemia.
Literature	GENERIF_SUMMARY	identified a coexpression of TF and MUC1 in the early secretory phase.
Literature	GENERIF_SUMMARY	the best immunohistochemical markers useful for the evaluation of tumors with chordoid morphology were D2-40, EMA, cytokeratin, and GFAP. D2-40 was a true chondroid marker to be useful for the differential diagnosis with chordoma
Literature	GENERIF_SUMMARY	findings suggest that KL-6 is a useful marker to assess the micropapillary character of breast and urinary bladder carcinomas.
Literature	GENERIF_SUMMARY	Results show that MUC1 is directly regulated by

Main Category	Category	Terms
		HIF-1alpha and affects the invasive and migration properties of renal cancer cells.
Literature	GENERIF_SUMMARY	Using the surface of the eye as a model system, Data found that galectin-3 colocalized with MUC1, on the apical surface of epithelial cells and that it bound to galectin-3 affinity columns in a galactose-dependent manner.
Literature	GENERIF_SUMMARY	Studies indicate that indicates that MUC1-C is a druggable target that is of potential importance to cancer treatment.
Literature	GENERIF_SUMMARY	constitutively activated STAT3 regulates expression of MUC1, which mediates lung cancer cell survival and metastasis in vitro and in vivo.
Literature	GENERIF_SUMMARY	expression of MUC1 and carbohydrate antigens in 124 tissue samples; 42 fibroadenoma (FA), 23 nonproliferative benign diseases (NPF), 25 usual epithelial hyperplasia (UEH), 7 atypical ductal hyperplasia (ADH), and 27 breast normal tissues.
Literature	GENERIF_SUMMARY	KL-6 mucin might be a specific tumor target for cholangiocarcinoma
Literature	GENERIF_SUMMARY	the expression of MUC1 might be associated with the progression of cholangiocarcinoma.
Literature	GENERIF_SUMMARY	the fate of MUC1 following the interaction with a ligand was studied.
Literature	GENERIF_SUMMARY	the interaction in the bloodstream of cancer patients between circulating galectin-3 and cancer cells expressing MUC1 bearing the galectin-3 ligand TF (Galbeta1,3GalNAc-) promotes metastasis
Literature	GENERIF_SUMMARY	a direct activator of NF-kappaB p65 and that an inhibitor of MUC1 function is effective in blocking activation of the NF-kappaB pathway
Literature	GENERIF_SUMMARY	MUC1 is a substrate for gamma-secretase
Literature	GENERIF_SUMMARY	breast cancer there was a limited humoral immune response through Lewis y/IgM/CIC levels detection which correlated with MUC1/IgM/CIC
Literature	GENERIF_SUMMARY	epithelial membrane antigen was focally expressed in only 8% of basal cell carcinoma
Literature	GENERIF_SUMMARY	Molecular evolution and specific fine-tuning thus have the potential to improve the performance of antibody specificities targeting tumor-associated epitopes on MUC1 mucin.
Literature	GENERIF_SUMMARY	MUC1 protects the epithelium from non-MUC1 binding bacteria by inhibiting adhesion to the cell surface by steric hindrance, and from MUC1-binding bacteria by acting as a releasable decoy.
Literature	GENERIF_SUMMARY	serum KL-6 is associated with the severity of RSV bronchiolitis and that it may be a useful biomarker for

Main Category	Category	Terms
		the severity of RSV bronchiolitis.
Literature	GENERIF_SUMMARY	we have provided evidence that may identify the MUC1 A/G polymorphism at 568 site, as a potential genetic factor which leads to an increase in susceptibility for gastric cancer
Literature	GENERIF_SUMMARY	MUC1 and STAT1 function in an auto-inductive loop, and activation of both MUC1 and the STAT1 pathway in breast tumors confers a poor prognosis for patients.
Literature	GENERIF_SUMMARY	Data provide evidence that some common variations in MUC1 and MUC5AC genes contribute to an elevated risk of stomach cancer.
Literature	GENERIF_SUMMARY	MUC1 might become a valuable adjunct to enable individual prognostic ramification prior to radical surgery in prostate cancer
Literature	GENERIF_SUMMARY	Serum KL-6 levels are elevated and appear to be correlated with pulmonary function variables in cystic fibrosis.
Literature	GENERIF_SUMMARY	Serum CA 15-3 is increased in pulmonary fibrosis.
Literature	GENERIF_SUMMARY	suppression of MUC1 by miR-145 causes a reduction of beta-catenin as well as the oncogenic cadherin 11
Literature	GENERIF_SUMMARY	MUC1 serves as a binding site for Pseudomonas aeruginosa on the surface of airway epithelial cells, which may have important consequences in the pathogenesis of Pseudomonas aeruginosa lung infections.
Literature	GENERIF_SUMMARY	Fine mapping of T-cell immunoglobulin mucin domain gene 1 failed to detect a significant association with multiple sclerosis.
Literature	GENERIF_SUMMARY	This is the first demonstration showing that MUC1 can suppress the virus-induced inflammatory responses.
Literature	GENERIF_SUMMARY	Thus, our data indicate that activation of the ERK pathway represents a new mechanism of Pem-mediated multiple resistances
Literature	GENERIF_SUMMARY	Down-regulation of ICAM-1 in mammary epithelial cells may contribute both to the high expression of psoriasin seen in some high-grade ductal carcinoma in situ and to the induction of MUC1.
Literature	GENERIF_SUMMARY	Data show that CEA and CA15.3 (CD227) are specific markers for diagnosis of early-stage breast cancer.
Literature	GENERIF_SUMMARY	Data show that plasma mammaglobin mRNA alone or in combination with CA15.3 (CD227) may be used as a valuable noninvasive approach for the diagnosis and the detection of metastasis in breast cancer at the time of diagnosis.
Literature	GENERIF_SUMMARY	High CA15-3 expression is associated with gallbladder carcinoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Results show that MUC1 is a useful molecular marker for the detection of malignant IPMN in pure pancreatic juice.
Literature	GENERIF_SUMMARY	tumor marker for breast cancer
Literature	GENERIF_SUMMARY	Benzalkonium chloride induces decreased expression of MUC1 in human conjunctival epithelial cells.
Literature	GENERIF_SUMMARY	regulates nuclear localization and function of the epidermal growth factor receptor
Literature	GENERIF_SUMMARY	Data show that Elevated CA 15-3 level is a useful parameter for predicting clinical outcomes.
Literature	GENERIF_SUMMARY	findings indicate that multiple myeloma cells are dependent on intact MUC1-C function for constitutive activation of the canonical NF-kappaB pathway and for their growth and survival
Literature	GENERIF_SUMMARY	Studies indicate that PPARgamma activation does not change PR binding to the MUC1 promoter but generally antagonizes progesterone activity by stimulating PRB degradation and inhibiting progesterone-induced PRB phosphorylation.
Literature	GENERIF_SUMMARY	The histological types of breast cancer are not significantly related to elevated serum levels of CEA and/or CA 15-3.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	results suggest that the interaction between free circulating galectin-3 and cancer-associated MUC1 promotes embolus formation and survival of disseminating tumour cells in the circulation
Literature	GENERIF_SUMMARY	Elevated Mucin-1 is associated with breast cancer.
Literature	GENERIF_SUMMARY	The expressions of MUC1 and/or MUC16 may be available as independent prognostic factors for the endocervical type of mucinous adenocarcinoma.
Literature	GENERIF_SUMMARY	major differences were seen in the ability of the constructs to bind shed and tumour cell-tethered MUC1. While the tribody did not discriminate, the bibody demonstrated preferential interaction with membrane-bound MUC1 compared with shed MUC1.
Literature	GENERIF_SUMMARY	Studies indicate that overexpression and aberrant glycosylation of MUC1 is associated with a wide variety of cancers, making it an ideal target for immunotherapeutic strategies.
Literature	GENERIF_SUMMARY	KL-6 has a role in bronchiolitis obliterans syndrome in lung transplant recipients
Literature	GENERIF_SUMMARY	By means of real-time quantitative polymerase chain reaction and immunohistochemical methods, 83 breast tissue samples were analyzed for MUC1 gene amplification and protein expression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	illustrate the capacity of MUC1 cytoplasmic tail to mediate transcription factor activity in a context-dependent manner to achieve wide spread and robust changes in gene expression
Literature	GENERIF_SUMMARY	A significant increase was found in serum IL-18, nitric oxide and CA 15.3 levels in sera of breast cancer patients when compared to those of the control group.
Literature	GENERIF_SUMMARY	These results demonstrate that Sjogren's syndrome subjects express greater quantities of MUC1 protein and mRNA compared to both keratoconjunctivitus sicca and control subjects.
Literature	GENERIF_SUMMARY	Expression of MUC1 subunits in the infertile endometrium is significantly different in fertile and appears to be a component of altered gene expression.
Literature	GENERIF_SUMMARY	MUC1 was expressed in 75.8% (147/194). MUC1 overexpression was detected in 50.0% (19/38 cases) dysplasia and 58.2% (32/55 cases) adenocarcinoma tissues.
Literature	GENERIF_SUMMARY	MUC1, MUC2, MUC5AC, and MUC6 may have a role in malignant transformation and invasion to mucosa or the muscularis mucosae in colorectal polyps
Literature	GENERIF_SUMMARY	The EMA was detected in all tumor cells except in the endothelial cells.
Literature	GENERIF_SUMMARY	These results support the idea about the involvement of MUC 5AC and MUC 1 with some specific sugar structures in the mechanism of Helicobacter pylori infection.
Literature	GENERIF_SUMMARY	MUC1-associated Tn/STn-epitopes are important targets for immunotherapy
Literature	GENERIF_SUMMARY	MUC1 mRNA and protein expression is increased by cytokine and P treatment in HES cells stably expressing PRB
Literature	GENERIF_SUMMARY	MUC1 expression may help differentiate follicular patterned thyroid lesions
Literature	GENERIF_SUMMARY	The MUC1 over-expression in renal clear-cell carcinoma(RCC) is associated with poor prognosis.
Literature	GENERIF_SUMMARY	MUC1 is the second major diffuse gastric cancer susceptibility gene identified. The SNPs rs2070803 and rs4072037 in MUC1 might be used to identify individuals at risk for this type of gastric cancer.
Literature	GENERIF_SUMMARY	higher expression of MUC 1 and MUC 4 is correlated with earlier stage mucoepidermoid carcinoma
Literature	GENERIF_SUMMARY	This study signifies the oncogenic role of MUC1 cytoplasmic tail and is the first to identify a direct role of the MUC1 in initiating EMT during pancreatic cancer.
Literature	GENERIF_SUMMARY	The immunohistochemical subtypes based on CK20 and MUC1 expression correlated with ampullary tumor

Main Category	Category	Terms
		progression.
Literature	GENERIF_SUMMARY	Strong MUC1 expression is associated with invasion in diffuse-type gastric carcinomas.
Literature	GENERIF_SUMMARY	Up-regulated expression of MUC1 indicated poorer overall survival and disease-free survival especially for those with non-squamous cell carcinomas.
Literature	GENERIF_SUMMARY	Studies indicate that MUC1, the most examined mucin family member in cutaneous pathologies, is normally expressed on the apical surface of glandular and ductal elements in several organs including the mammary gland, lungs and the pancreas.
Literature	GENERIF_SUMMARY	Results suggest that CLEC-muc is a novel stem cell-like population with CD227 positivity and unique pattern of P63 expression that can be further differentiated by BMP4 to generate specific cell-types probably destined to form non-keratinized epithelia.
Literature	GENERIF_SUMMARY	study found that mucins, including MUC-1 derived from rheumatoid arthritis synovial fluid, induced IL-6 production on PBMCs and that MUC-1 was expressed on synovial cells and mononuclear cells in RA synovial tissues
Literature	GENERIF_SUMMARY	expression of MUC1 was closely related to the grade of malignancy in thymic epithelial tumors. A positive expression of MUC1 was significantly associated with poor outcome in this disease
Literature	GENERIF_SUMMARY	Galectin-3 regulates MUC1 and epidermal growth factor receptor (EGFR) internalization and subcellular localization, ERK1,2 activation downstream of the EGFR and EGFR nuclear translocation in pancreatic cancer cells.
Literature	GENERIF_SUMMARY	Androgen receptor signaling regulates MUC1 expression by transcriptional and posttranscriptional mechanisms in prostate cancer cells.
Literature	GENERIF_SUMMARY	Increased mucin 1 expression in recurrence and malignant transformation of salivary gland pleomorphic adenoma
Literature	GENERIF_SUMMARY	Data show that MUC1-C interacts directly with JAK1 and STAT3.
Literature	GENERIF_SUMMARY	findings indicate that MUC1 contributes to ESCC metastasis by stimulating MMP13 expression, suggesting MUC1 as a novel diagnostic biomarker and therapeutic target in ESCC
Literature	GENERIF_SUMMARY	Data suggest that preoperative serum KL-6 level reflects KL-6 expression patterns in NSCLC tissue, and can serve as a useful prognostic biomarker in NSCLC patients who undergo curative surgery.
Literature	GENERIF_SUMMARY	MUC1 and Gal-1 might be useful for evaluating the potential of leiomyoma to transform into leiomyosarcom.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Sustained localization of KL-6 mucin in the surrounding membrane and/or cytoplasm of cancer cells at the invasion front has a significant role in cancer progression.
Literature	GENERIF_SUMMARY	Epithelial lining fluid and serum KL-6 concentrations were found to be good indicators of clinical outcome in acute respiratory distress syndrome patients.
Literature	GENERIF_SUMMARY	NSCLC cells are dependent on MUC1-C both for activation of the PI3K --> Akt pathway and for survival.
Literature	GENERIF_SUMMARY	MUC1-C subunitfunction is of importance to acute myelogenous leukemia cell self-renewal
Literature	GENERIF_SUMMARY	Data suggest that attention should be paid to the emergence of invasive IPMN from flat foci in MUC2-positive and MUC1/2-negative cases.
Literature	GENERIF_SUMMARY	High MUC1 is associated with platinum-resistant ovarian cancer.
Literature	GENERIF_SUMMARY	This study indicated that KL-6/MUC1 played a complex role in regulating pancreatic ductal carcinoma cell proliferation, apoptosis, and invasion; there may be a functional link between KL-6/MUC1 expression and beta-catenin subcellular localization
Literature	GENERIF_SUMMARY	Polymorphisms in MUC1, MUC2, MUC5B and MUC6 genes are not associated with the risk of chronic atrophic gastritis.
Literature	GENERIF_SUMMARY	Findings indicated that MUC1 contributes to cell-cell adhesion.
Literature	GENERIF_SUMMARY	The expression of MUC1 and MUC2 was not significantly higher in neoplastic than nonneoplastic tissues.
Literature	GENERIF_SUMMARY	Mucin and MUC5AC expression is selectively induced by Aspergillus fumigatus extracts or the culture supernatant of live microorganisms in NCI-H292 tumor cells and primary cultures of human airway epithelial cells.
Literature	GENERIF_SUMMARY	MUC1 and MUC2 expression was useful for classification of ampullary cancer.
Literature	GENERIF_SUMMARY	the beta subunits of MUC1 and MUC4 have emerged as oncogenes [review]
Literature	GENERIF_SUMMARY	Frequent EMA expression in EFH represents an unexpected finding and constitutes a potential diagnostic pitfall.
Literature	GENERIF_SUMMARY	Results of this study strongly support the potential of KL-6 as a diagnostic biomarker for life-threatening EGFR tyrosine kinase inhibitor induced interstitial lung disease.
Literature	GENERIF_SUMMARY	The serum levels of CA 15-3 were independent of trimester stage.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Non-cysteine linked MUC1 dimerization in cell signalling pathways required for cancer cell migration.
Literature	GENERIF_SUMMARY	Placental mucin 1 staining was reduced and serum concentrations of mucuin 1 were increased in cases of fetomaternal hemorrhage.
Literature	GENERIF_SUMMARY	High MUC1 is associated with gallbladder adenocarcinoma.
Literature	GENERIF_SUMMARY	study characterised the glycosylation patterns of MUC1 and CEACAM5 in colorectal cancer patients comparing normal colon and tumour epithelium within each patient
Literature	GENERIF_SUMMARY	High serum KL-6 is associated with lung cancer.
Literature	GENERIF_SUMMARY	Our findings indicate that MUC1 may play a role in the regulation of hormone receptors in invasive ductal breast carcinomas
Literature	GENERIF_SUMMARY	MUC1 is overexpressed in severe preeclamptic placentas and that MUC1 overexpression suppresses extravillous trophoblast invasion mainly via modulating beta1-integrin signaling.
Literature	GENERIF_SUMMARY	The current report is the first to demonstrate that MUC1 expression promotes angiogenesis in human breast cancer in vivo and in vitro.
Literature	GENERIF_SUMMARY	The data strongly suggest that the expression levels of MUC1 may be used as a diagnostic test in dry eye syndrome for investigational and selective clinical trials.
Literature	GENERIF_SUMMARY	The administration of MUC1-peptide-pulsed dendritic cells is non-toxic and capable of inducing immunological response to tumor antigen MUC1 in advanced pancreatic cancer patients.
Literature	GENERIF_SUMMARY	Overexpression of MUC1 enhances proangiogenic activity of non-small-cell lung cancer cells through activation of Akt and extracellular signal-regulated kinase pathways.
Literature	GENERIF_SUMMARY	Low MUC1 expression is associated with pulmonary metastatic tumors.
Literature	GENERIF_SUMMARY	Data report that levels of MUC1 are significantly higher in subpopulations of cells with elevated intrinsic Deltapsim derived from both mammary and colonic carcinoma cell lines.
Literature	GENERIF_SUMMARY	MUC1 may be involved in human nephrogenesis and may play a relevant role in mesenchymal-epithelial transition from the cap mesenchyme to the renal vesicle, changing the fate of renal stem/progenitor cells.
Literature	GENERIF_SUMMARY	Gastrointestinal stromal tumours infrequently show focal EMA positivity
Literature	GENERIF_SUMMARY	MUC1 protein expression in tumor cells regulates transcription of proinflammatory cytokines by forming a

Main Category	Category	Terms
		complex with nuclear factor-kappaB p65 and binding to cytokine promoters
Literature	GENERIF_SUMMARY	In patients with operable breast cancer, presurgical CA15.3 value is an independent prognostic factor for metastases and deaths.
Literature	GENERIF_SUMMARY	results indicate that MUC1 downregulation alters the phenotype and tumorigenicity of MKN45 gastric carcinoma cells and also the expression of several molecules that can be involved in tumorigenic events
Literature	GENERIF_SUMMARY	MUC1 is expressed on the majority of T-regulatory cells (CD4(+)/CD25(+)/FoxP3(+)) in humans (>90%).
Literature	GENERIF_SUMMARY	Depolarized MUC1 expression was significantly associated with poor outcome in patients with lung cancer.
Literature	GENERIF_SUMMARY	In epithelioid mesotheliomas, 13% of cells expressed nestin, 39% EMA and 7% mesothelin
Literature	GENERIF_SUMMARY	In colon cancer, the expression of mucin1 can be promoted by rIL-6, while the expression of mucin2 can be inhibited.
Literature	GENERIF_SUMMARY	Forty-seven specimens of renal parenchyma tumors were studied immunohistochemically for the expression of EMA.
Literature	GENERIF_SUMMARY	A serum KL-6 level of higher than 500 U/mL is a marker of the presence of interstitial pneumonitis, and a level of higher than 1,000 U/mL is a marker of the presence of active interstitial pneumonitis associated with connective tissue diseases.
Literature	GENERIF_SUMMARY	These findings indicate that the MUC1-C subunit regulates glycolysis and that this response is conferred in part by PKM2.
Literature	GENERIF_SUMMARY	MUC1 expression by MPM cells, and its recognition by a MUC1-specific CD8(+) T-cell clone, was downregulated by HDACi when used alone or in combination with 5-azaCdR.
Literature	GENERIF_SUMMARY	MUC1-C cytoplasmic domain cysteines are necessary and sufficient for MUC1-C dimerization and may be targeted by antineoplastic agents.
Literature	GENERIF_SUMMARY	expression of MUC1 at the cell surface is also observed and found to be expressed only after HRG stimulation. The total amount of MUC1 remains unchanged
Literature	GENERIF_SUMMARY	human airway epithelia express catalytically active NEU1 sialidase that regulates EGFR- and MUC1-dependent signaling and bacterial adhesion.
Literature	GENERIF_SUMMARY	epidermal growth factor receptor tyrosine phosphorylates MUC1, leading to an increase in its association with TLR5, thereby competitively and reversibly inhibiting recruitment of MyD88 to TLR5 and downstream signaling events.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The levels of sMUC-1 in patients with multiple myeloma
Literature	GENERIF_SUMMARY	MUC1-induced invasion and proliferation occurs via increased exogenous production of PDGFA in pancreatic ductal carcinoma.
Literature	GENERIF_SUMMARY	data are consistent with the hypothesis PPARgamma stimulates MUC1/Muc1 expression, thereby blocking phorbol 12-myristate 13-acetate-induced TNF-alpha/ IL-8 production by airway epithelial cells.
Literature	GENERIF_SUMMARY	Similar MUC1, MUC2 and MUC5B expression patterns were observed between mucocele-like lesions and mucinous carcinoma (MC) type A and between MC type B and mucinous carcinoma.
Literature	GENERIF_SUMMARY	This report is the first, to the best of our knowledge, demonstrating that the inflammatory response in airway epithelial cells during infection with NTHi is controlled by MUC1 mucin, mainly through the suppression of TLR2 signaling
Literature	GENERIF_SUMMARY	We conclude that MUC1 does not seem to be an adequate marker for differential diagnostics of clear cell renal carcinoma metastatic to the pancreas
Literature	GENERIF_SUMMARY	MUC1-C oncoprotein contributes to TCF7L2 activation and thereby promotes cyclin D1 expression in breast cancer cells.
Literature	GENERIF_SUMMARY	the MUC1 aptamer selectively delivers cytotoxic agent to cancer cells in vitro
Literature	GENERIF_SUMMARY	No significant association was found with variants in the MUC1 gene and evolution of gastric cancer precursor lesions.
Literature	GENERIF_SUMMARY	Compared serum sFas and p53 protein with that of serum CA 15-3 as the most commonly used breast cancer tumor marker. Serum samples were from breast cancer patients before surgery, 2 weeks after surgery and after six cycles of FAC chemotherapy.
Literature	GENERIF_SUMMARY	The pathological expression of activin beta A, iNOS and MUC1 by fallopian tubes bearing an ectopic pregnancy is not involved in the determination of implantation site.
Literature	GENERIF_SUMMARY	High CA 15-3 is associated with hormone receptor 2 positive breast cancer.
Literature	GENERIF_SUMMARY	MUC1 carrying core 2 O-glycans thus functions as a molecular shield against natural killer cell attack, thereby promoting bladder tumor metastasis.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	MUC1-C oncoprotein confers androgen-independent growth of human prostate cancer cells.
Literature	GENERIF_SUMMARY	MUC1-C oncoprotein binds directly to the BAX BH3 domain and thereby blocks BAX function in activating the mitochondrial death pathway.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	SF-1 and EMA may be helpful in the differentiation of adrenal tumors from renal tumors in difficult cases.
Literature	GENERIF_SUMMARY	The binding characteristics outlined here suggest that the interaction between Src-SH3 and MUC1-CD represents a novel weak electrostatic interaction of the type which is increasingly recognized as important in transient and dynamic protein complexes required for cell migration and signal transduction.
Literature	GENERIF_SUMMARY	It is currently one of the best and most reliable serum biomarkers available for interstitial lung diseases.
Literature	GENERIF_SUMMARY	EGFR and Src activity contribute to carcinoma cell invasion and metastasis mediated by integrin alphavbeta5 in part by promoting proteolytic cleavage of MUC1 .
Literature	GENERIF_SUMMARY	significantly elevated CA15-3 serum levels were found among grade II, III and stage II and III breast cancer females compared to normal healthy females; elevated CA15-3 serum levels were also found among ER /PR and ER /PR- compared to healthy control females
Literature	GENERIF_SUMMARY	The molecular structural variants of KL-6/MUC1 and its leakage behavior affect serum levels of KL-6 in sarcoidosis.
Literature	GENERIF_SUMMARY	Regulation of MUC1-C expression is mediated by the PI3K/AKT pathway and the eIF4A RNA helicase.
Literature	GENERIF_SUMMARY	Administration of anti-hMUC1 IgE reduced the growth of MUC1(+) tumors.
Literature	GENERIF_SUMMARY	Mucin-1 expression is statisically significantly reduced in women with recurrent pregnancy loss.
Literature	GENERIF_SUMMARY	MUC-1 protein, human blood level is probably a good indicator for the the prognosis in Duchenne dystrophy.
Literature	GENERIF_SUMMARY	These results support the hypothesis that PPARgamma stimulates MUC1 expression by AGS cells, thereby attenuating H. pylori- and PMA-induced IL-8 production.
Literature	GENERIF_SUMMARY	MUC1-C and Rab31 function in an autoinductive loop that contributes to overexpression of MUC1-C in breast cancer cells.
Literature	GENERIF_SUMMARY	MUC1 and MSLN were superior to pathologic features and other putative biomarkers as predicting survival group.
Literature	GENERIF_SUMMARY	rs4072037 polymorphisms in MUC1 is associated with esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Intraductal tubulopapillary neoplasm of the pancreas cells exhibited luminal immunolabeling for MUC-1.
Literature	GENERIF_SUMMARY	Identify MUC1-CT and p120ctn as important regulators of epithelial polarity and cell-cell adhesion during a smoke-induced epithelial mesenchymal transition-like process.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	removing MUC1 from decidual tissue might help control trophoblast invasion by NK cells.
Literature	GENERIF_SUMMARY	MUC1 mucin stabilizes and activates hypoxia-inducible factor 1 alpha to regulate metabolism in pancreatic cancer
Literature	GENERIF_SUMMARY	MUC1-N translocates to the nucleus where it is expressed in nuclear speckles and MUC1-N and MUC1-C have dissimilar intranuclear distribution patterns
Literature	GENERIF_SUMMARY	study suggested rs2294008 in the PSCA gene to be associated with increased risk of gastric cancer and rs2070803 in MUC1 to play a protective role in a Chinese population
Literature	GENERIF_SUMMARY	We now report that one new G SNP-associated isoform MUC1/Y-LSP, but not the A SNP-associated isoform MUC1/Y, inhibits tumor growth in immunocompetent but not immunocompromised mice.
Literature	GENERIF_SUMMARY	the crosstalk between MUC1 and c-Met in HCC could provide an advantage for invasion to HCC cells through the beta-catenin/c-Myc pathway.
Literature	GENERIF_SUMMARY	novel MUC1 functions in ovarian cancer biology and support its suitability as a target for immune-based therapies.
Literature	GENERIF_SUMMARY	Increases in serum KL-6 and SP-D levels during the first 4 weeks after starting therapy, but not their levels at any one time point, predict poor prognosis in patients with polymyositis/dermatomyositis.
Literature	GENERIF_SUMMARY	The results show that single nucleotide polymorphism in the MUC1 gene affect serum levels of KL-6 in both German and Japanese populations
Literature	GENERIF_SUMMARY	A risk score model derived from a logistic regression analysis of Glut-1 and EMA immunocytochemistry greatly improves the distinction between malignant mesothelioma from reactive mesothelial hyperplasia cells in individual effusions.
Literature	GENERIF_SUMMARY	tumor delta-T2s were significantly reduced by treatment with Dox indicating lower accumulation of MN-EPPT. This correlated with a lower level of MUC-1 expression in the Dox-treated tumors, as confirmed by immunoblotting
Literature	GENERIF_SUMMARY	Data suggest that expression of cell membrane-associated mucins MUC1, MUC4, and MUC16 are differentially regulated in endocervical epithelial cell aggregates by different bacterial and viral products (here, toll-like receptor ligands/agonists).
Literature	GENERIF_SUMMARY	MUC1 and CFTR mRNA and protein expression decrease in the endometrium of infertile hydrosalpinx patients.
Literature	GENERIF_SUMMARY	MUC1 has a pivotal role in controlling hypoxia-driven angiogenesis via the regulation of multiple

Main Category	Category	Terms
		proangiogenic factors in pancreatic cancer.
Literature	GENERIF_SUMMARY	MUC1 overexpression and underglycosylation in breast cancer is associated with tumor invasiveness and metastatic potential.
Literature	GENERIF_SUMMARY	The expression of MUC4 as well as MUC1 in early gastric cancers is a useful marker to predict poor prognostic factors related with vessel invasion.
Literature	GENERIF_SUMMARY	Overexpression of EZH2 may be associated with malignant behaviour in intraductal papillary neoplasm of the bile duct in parallel with up-regulated MUC1 expression and down-regulated MUC6 expression.
Literature	GENERIF_SUMMARY	MUC1 strongly promoted all of these events.
Literature	GENERIF_SUMMARY	High serum CA 15-3 level is associated with breast cancer.
Literature	GENERIF_SUMMARY	Peak serum KL-6 level >/= 700 U/ml may become a clinically useful marker of poor prognosis for diffuse alveolar hemorrhage.
Literature	GENERIF_SUMMARY	Sialylation of the MUC1 tandem repeating glycopeptide, Pro-Pro-Ala-His-Gly-Val-Thr-Ser-Ala-Pro-Asp-Thr-Arg-Pro-Ala-Pro-Gly-Ser-Thr-Ala with core 2-type O-glycans, afforded a specific conformational change at one of the cancer-relevant epitopes.
Literature	GENERIF_SUMMARY	CA 15-3 has been identified as a valuable marker for pretherapeutic stratification of breast cancer patients with liver metastasis and early estimation of response to selective internal radiation therapy
Literature	GENERIF_SUMMARY	ALKBH2 is an upstream molecule of the oncoprotein, MUC1, and regulates cell cycle and EMT, resulting in progression of urothelial carcinomas.
Literature	GENERIF_SUMMARY	Serum levels of leptin, adiponectin and CA 15-3, as well as anthropometric and biochemical parameters were analysed in 88 women were diagnosed with either breast cancer (n = 40) or benign breast lesions (n = 48).
Literature	GENERIF_SUMMARY	can be used as a tumor marker for symptomatic metastasis in breast cancer patients
Literature	GENERIF_SUMMARY	Studies indicate the diverse roles of mucin-type O-glycosylation during eukaryotic development.
Literature	GENERIF_SUMMARY	MUC1-shRNA could inhibit proliferation and significantly weaken HIF-1alpha protein expression of QBC939 cells, suggesting its potential as a therapeutic target of CCA.
Literature	GENERIF_SUMMARY	MUC1 expression characterizes multiple structures during human nephrogenesis.
Literature	GENERIF_SUMMARY	Respiratory syncytial virus dthe expression of MUC1 in airway epithelial cells whereas influenza virus could not.
Literature	GENERIF_SUMMARY	We considered the possibility that medullary cystic

Main Category	Category	Terms
		kidney disease type 1 might be caused by point mutations in the MUC1 gene
Literature	GENERIF_SUMMARY	KL-6 levels tended to increase after interstitial pneumonitis was detected on imaging in pegylated interferon treated hepatitis C patients.
Literature	GENERIF_SUMMARY	Data indicate that histologic subtype and CDX2 and MUC1 expression were significant prognostic variables.
Literature	GENERIF_SUMMARY	Data indicate that MUC1 and MUC4 were differentially glycosylated as the disease progressed, and the expression of MUC6 in acinar cells.
Literature	GENERIF_SUMMARY	The MUC1-associated sLe(a) is generated in these cells after introduction of MUC1 cDNA.
Literature	GENERIF_SUMMARY	MUC1-C contributes to tamoxifen resistance.
Literature	GENERIF_SUMMARY	MUC1 (CD227) expression was associated with infiltration of CD10 myeloid cells, TGF-beta1 expression, and tumor budding grade.
Literature	GENERIF_SUMMARY	cells expressing MUC1 exhibited decreased cell proliferation, increased cell cycle arrest and reduced cell migratory and invasive capacities.
Literature	GENERIF_SUMMARY	MUC1 and HER2 might be closely associated with invasive phenotype of intraductal papillary mucinous neoplasm.
Literature	GENERIF_SUMMARY	High-grade DCIS or IC showed increasing loss of apical polarised and cytoplasmic expression of MUC1.
Literature	GENERIF_SUMMARY	In multivariable analysis, MUC1 protein and mRNA expression were independently predictive for overall survival.
Literature	GENERIF_SUMMARY	High serum CA15-3 levels are associated with metastatic breast cancer.
Literature	GENERIF_SUMMARY	The expression of mucin 1, cell surface associated (MUC1) and mucin 16, cell surface associated (MUC16) proteins and messenger ribonucleic acid (mRNA) in a cohort of postmenopausal women, was quantified.
Literature	GENERIF_SUMMARY	Ca 15.3 serum levels strongly correlated with radiology scores for interstitial disease in systemic sclerosis patients.
Literature	GENERIF_SUMMARY	The increased expression of MUC1 and the Lewis y antigen is a significant risk factor for chemoresistance in patients with ovarian epithelial cancer.
Literature	GENERIF_SUMMARY	The existence of IgG autoantibodies against aberrantly glycosylated MUC1 may have a protective role and may contribute to a better prognosis in some breast cancer patients.
Literature	GENERIF_SUMMARY	The involvement of Lewis b and H type 1 structures of H. pylori in adhesion of bacteria to MUC1 mucin, was investigated.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	MUC1 is expressed at high frequency in early-stage basal-like triple-negative breast cancer.
Literature	GENERIF_SUMMARY	MUC1 immunoreactivity was observed in effusion fluids from 99.0%of cases of metastatic adenocarcinoma and in 48.8% of cases of reactive mesothelial cells
Literature	GENERIF_SUMMARY	High MUC1 expression is associated with acute myeloid leukemia.
Literature	GENERIF_SUMMARY	Downregulation of MUC1 increases tumor-promoting myeloid-derived suppressor cells.
Literature	GENERIF_SUMMARY	Enhancement of in vitro cell motility and invasiveness of human malignant pleural mesothelioma cells through the HIF-1alpha-MUC1 pathway.
Literature	GENERIF_SUMMARY	High CA15-3 exprwession is associated with recurrence in breast cancer.
Literature	GENERIF_SUMMARY	MUC1-C contributes to constitutive activation of the HER2 pathway and targeting MUC1-C represents a potential approach to abrogate trastuzumab resistance.
Literature	GENERIF_SUMMARY	Mucin expression was not predictive of overall survival benefit from adjuvant chemotherapy in non-small cell lung cancer patients.
Literature	GENERIF_SUMMARY	Data show that prior mastitis was associated with a significantly higher anti-CA15.3 and anti-CA125 antibody levels and lower risk of ovarian cancer.
Literature	GENERIF_SUMMARY	The results of mucin immunohistochemical staining of preoperative biopsy and surgically resected specimens were in agreement with each other, and in close agreement with those for pancreatic juice cytology
Literature	GENERIF_SUMMARY	Correlation between MUC1 and MUC16 expression and antibody dependent cellular-cytotoxicity and cytotoxic T-lymphocyte mediated cell killing.
Literature	GENERIF_SUMMARY	MUC1-C (i) activates a previously unrecognized ERK-->C/EBPbeta-->ALDH1A1 pathway, and (ii) promotes the induction of ALDH activity in breast cancer cells.
Literature	GENERIF_SUMMARY	MUC1 hypermethylation is associated with triple-negative breast cancers.
Literature	GENERIF_SUMMARY	Siglec-9 expressed on immune cells may play a role as a potential counterreceptor for MUC1 and that this signaling may be another MUC1-mediated pathway and function in parallel with a growth factor-dependent pathway.
Literature	GENERIF_SUMMARY	MUC1 directly binds to JNK1 through the MUC1-CD (amino acids 1-45) region.
Literature	GENERIF_SUMMARY	the detection of EpCAM/MUC1 mRNA-positive CTCs in the blood before and after surgery is useful for predicting a poor prognosis in NSCLC patients who undergo curative surgery.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Elevated serum CA 15-3 and CEA levels at recurrence suggest increased tumor burden and may be prognostic for survival for metastatic breast cancer patients.
Literature	GENERIF_SUMMARY	These findings represent important evidence for association of MUC1 rs4072037 variant with gastric cancer risk.
Literature	GENERIF_SUMMARY	Participation of Lewis b, H type 1 and also sialyl Lewis x of MUC1 mucin in interactions with H. pylori independently, is reported.
Literature	GENERIF_SUMMARY	Prominent EMA dots in Bergmann glia might represent excessive activation induced by an overlying leptomeningeal tumour that stimulates the expression of early developmental antigens.
Literature	GENERIF_SUMMARY	signaling through MUC1 influences cancer progression by regulating transcription of microRNAs that are associated with the process of metastasis
Literature	GENERIF_SUMMARY	this study suggests that miR-145 modulates ovarian cancer growth and invasion by suppressing p70S6K1 and MUC1, functioning as a tumor suppressor.
Literature	GENERIF_SUMMARY	A novel platform based on a hairpin oligonucleotide (HO) switch, gold nanoparticles (AuNPs), and enzyme signal amplification for the ultrasensitive detection of mucin 1 protein (MUC1) was developed in this assay
Literature	GENERIF_SUMMARY	Overall, based on the findings SPIONs-C595 nanosized probe is a selective ovarian molecular imaging modality.
Literature	GENERIF_SUMMARY	Our findings indicate that aberrant glycosylation of MUC1 is an early event in the pathogenesis of multiple myeloma. In contrast, MUC1C nuclear localization is not likely to be a driver of tumour progression.
Literature	GENERIF_SUMMARY	aim of this study was to determine whether rs4072037A > G in MUC1 at 1q22 and rs2274223A > G in PLCE1 at 10q23 are associated with a risk of gastric cancer in a Korean population
Literature	GENERIF_SUMMARY	The MUC1-C oncoprotein is aberrantly expressed in AML cells and contributes to activation of the mutant FLT3 receptor.
Literature	GENERIF_SUMMARY	MUC1 expression in macrophages has a role in cigarette smoke-induced lung cancer through a mechanism involving the nuclear receptor PPAR-gamma
Literature	GENERIF_SUMMARY	PPARgamma induces MUC1-C ubiquitination and degradation that is critical to terminate MUC1-C signaling pathway-elicited cancer
Literature	GENERIF_SUMMARY	P120 catenin ARM domains 1, 3-5, and 8 mediate interactions between p120 catenin and MUC1.
Literature	GENERIF_SUMMARY	MUC1 and SNAIL were overexpressed in human renal sarcomatoid carcinomas.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	tumorigenicity experiments showed that inhibition of Muc1 could significantly suppress the growth of HCC2998 in nude mice models
Literature	GENERIF_SUMMARY	This article provides a comprehensive review of recent literature on novel metabolic functions of MUC1
Literature	GENERIF_SUMMARY	Adolescents with very high intensity of dental caries disease had increased levels of MUC1 and MUC5B.
Literature	GENERIF_SUMMARY	Polymorphisms in MUC1 gene are not associated with helicobacter pylori infection and non-cardia gastric cancer.
Literature	GENERIF_SUMMARY	A high MUC1 expression was associated with a favourable prognosis in patients with bladder cancer when the expression of HER3 was also high. This suggests an involvement of HER3 in MUC1 function in bladder cancer.
Literature	GENERIF_SUMMARY	Data suggest that PAM4-reactivity for MUC1 can sever as diagnostic tumor marker to discriminate pancreatic ductal adenocarcinoma (and neoplastic precursor lesions) from benign pancreatic tissues, particularly inflamed chronic pancreatitis.
Literature	GENERIF_SUMMARY	Data indicate that Helicobacter pylori decreases methylation of the MUC1 gene.
Literature	GENERIF_SUMMARY	MUC1 mutation results in progressive chronic kidney failure with a bland urinary sediment in medullary cystic kidney disease type 1.
Literature	GENERIF_SUMMARY	Seven polymorphisms of MUC16 and MUC1 genes failed to identify and confirm effects on ovarian cancer risk development.
Literature	GENERIF_SUMMARY	CDX2, MUC1, MUC5AC, and MUC6 expression patterns may provide better estimations of survival after surgical resection of small intestinal adenocarcinoma.
Literature	GENERIF_SUMMARY	Data show targeting MUC1-C is synergistic with bortezomib (BTZ) in suppressing p53-inducible regulator of glycolysis and apoptosis (TIGAR)-mediated regulation of reactive oxygen species levels for combining GO-203 with BTZ in BTZ resistance.
Literature	GENERIF_SUMMARY	The serological levels of CEA and CA 15-3 in breast cancer patients were significantly higher than the serum levels of normal controls.
Literature	GENERIF_SUMMARY	Tumor-associated MUC1 differs from the MUC1 expressed in normal cells with regard to its biochemical features, cellular distribution, and function.
Literature	GENERIF_SUMMARY	The expression of MUC1 is increased while that of MUC4 decreased in urothelial carcinoma compared to the normal non-neoplastic urothelium.
Literature	GENERIF_SUMMARY	Higher preoperative CA 15-3 is associated with worse clinical outcomes exclusively in patients with nonmetastatic luminal A breast cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	MUC1 has a role in regulating epithelial inflammation and apoptosis by PolyI
Literature	GENERIF_SUMMARY	In rare adenosquamous carcinoma of the head and neck region, it was revealed that MUC1 and MUC4 expression might be associated with its nodal status.
Literature	GENERIF_SUMMARY	The results indicated that the DNA methylation status of MUC1, MUC2 and MUC4 in pancreatic juice matched with the mucin expression in tissue.
Literature	GENERIF_SUMMARY	positively relates to nasal polyp recurrences
Literature	GENERIF_SUMMARY	MUC1 splice variants are correlated with tumor progression in esophageal cancer.
Literature	GENERIF_SUMMARY	Collectively, our results indicate that MUC1 plays a key role in HCC tumorigenesis.
Literature	GENERIF_SUMMARY	MUC1 rs4072037 polymorphism may be used as potential biomarker for cancer susceptibility particularly for gastric cancer and for Asian population.
Literature	GENERIF_SUMMARY	High MUC1 expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	KL-6 can be used as a lung fibrosis severity marker in systemic sclerosis, but its role as a disease activity marker is questionable. Following cyclophosphamide treatment serum KL-6 levels may decrease independently of the lung function parameters.
Literature	GENERIF_SUMMARY	Aberrant expression of CD227, especially cytoplasm staining could be predictive for tumor aggressiveness, lymph node metastasis, poorer outcome of patients with breast cancers.
Literature	GENERIF_SUMMARY	MUC1 is known as an oncogene with an anti-apoptotic function in cancer cells; in normal gastric mucosa, MUC1 has a role in protecting gastric epithelial cells from a variety of external insults which cause inflammation and carcinogenesis.
Literature	GENERIF_SUMMARY	cancer antigen 15.3 may be correlated with results of positron emission tomography/computed tomography in breast cancer patients
Literature	GENERIF_SUMMARY	Aberrant glycosylation of MUC1-N in response to cigarette smoke plays a functional role in the initiation of epithelial-mesenchymal transition (EMT), thereby serving as an early indicator of malignant transformation.
Literature	GENERIF_SUMMARY	GCNT1 is over-expressed in prostate cancer and is associated with higher levels of core 2 O-sLe(x) in PSA, PAP and MUC1 proteins.
Literature	GENERIF_SUMMARY	These results indicate that CA15-3, CA125, CEA, and TSGF in nipple discharge can serve as novel biomarkers in the diagnosis and prognosis of breast cancer.
Literature	GENERIF_SUMMARY	In idiopathic pleuroparenchymal fibroelastosis SP-D

Main Category	Category	Terms
		was elevated, while KL-6 was within a normal range.
Literature	GENERIF_SUMMARY	Serial increases in CA15-3 can be used to predict positive PET/CT results in breast cancer patients during follow-up.
Literature	GENERIF_SUMMARY	First degree relatives of gastric cancer patients show increased expression of MUC1, which may serve as a predictor of future intestinal metaplasia, dysplasia, and cancer.
Literature	GENERIF_SUMMARY	The KL-6 antibody is immunohistochemically a good biomarker of the PB subtype of IPMC.
Literature	GENERIF_SUMMARY	The genetic length variants in the central repetitive region of MUC1 were not associated with meconium ileus in cystic fibrosis.
Literature	GENERIF_SUMMARY	MUC1, MUC4, and MUC16 expression does not vary significantly in normal endometrium during the menstrual cycle or in endometriosis relative to the epithelial marker, cytokeratin-18 (KRT18).
Literature	GENERIF_SUMMARY	role of MUC1 and MUC16 in epithelial barrier function
Literature	GENERIF_SUMMARY	Our study shows an association between MUC1 expression and the presence of the BRAF(V600E) mutation in papillary thyroid carcinoma.
Literature	GENERIF_SUMMARY	The meta-analysis indicates that Mucin-1 (MUC1) assay plays an important role in the diagnosis of pancreatic ductal adenocarcinoma (PDAC).
Literature	GENERIF_SUMMARY	the electrodepositing gold nanoparticles (AuNPs) on the bare glass carbon electrode (GCE) with enhanced surface area could capture a large amount of primary anti-CA15-3 to improve the sensitivity of the immunosensor
Literature	GENERIF_SUMMARY	Beas-2B cells exposed to Ni(2+) showed increased expression of MUC1 levels. This correlated with increased phosphorylation of both EGFR and of the ERK1/2 and increased resistance to apoptosis.
Literature	GENERIF_SUMMARY	Fallopian tubes with hydrosalpinx have a selective accumulation of MUC1 cytoplasmic tail, but not difference in the ectodomain.
Literature	GENERIF_SUMMARY	Elevated levels of epithelial membrane antigen is associated with breast cancer.
Literature	GENERIF_SUMMARY	our results establish a pathway consisting of miR-551b/catalase/ROS that results in MUC1 overexpression, and intervention against this pathway could be exploited to overcome acquired chemoresistance.
Literature	GENERIF_SUMMARY	potential markers distinguishing normal renal tissue from intermediate zone and from cancer renal cells during clear cell renal cell carcinoma development
Literature	GENERIF_SUMMARY	Baseline levels of KL-6 were not statistically significantly different between rheumatoid arthritis

Main Category	Category	Terms
		patients with and without interstitial lung disease events, but those levels increased in the former patient group at the time of ILD events.
Literature	GENERIF_SUMMARY	Expression of CCN2, EMA, and FAP may be involved in the activation of cancer associated fibroblasts in hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Decreased expression of MUC1 may attribute to progesterone receptor B variation in the mid-secretory phase of endometriosis endometrium; it might be one of the factors resulting in infertility in endometriosis patients
Literature	GENERIF_SUMMARY	The reverse polarity of colorectal micropapillary carcinomas is sometimes better visualised by immunostains other than EMA/MUC1.
Literature	GENERIF_SUMMARY	Mucin 1 downregulation associates with corticosteroid resistance in chronic rhinosinusitis with nasal polyps.
Literature	GENERIF_SUMMARY	An increased expression of MUC1 may be of great value in assessing the development and prognosis of salivary gland mucoepidermoid carcinoma, and could be used as a new molecule target to improve outcomes for these patients in the future.
Literature	GENERIF_SUMMARY	Targeting MUC1-C inhibits mutant EGFR signaling and survival.
Literature	GENERIF_SUMMARY	KL-6/MUC1 glycosylation is involved in pancreatic cancer metastasis and invasion.
Literature	GENERIF_SUMMARY	These findings support a model in which targeting MUC1-C inhibits mutant KRAS signaling in Non-small cell lung cancer cells and thereby reverses the epithelial-mesenchymal transition phenotype and decreases self-renewal.
Literature	GENERIF_SUMMARY	Overexpression of MUC1 prevents initiation of anoikis in epithelial cancer cells in response to loss of adhesion.
Literature	GENERIF_SUMMARY	MUC1 expression in ovarian mucinous tumors did not change from benign to malignant tumors.
Literature	GENERIF_SUMMARY	Suggest that feedback activation of STAT3 via up-regulation of MUC1/4 expression constitutes a key node mediating trastuzumab resistance.
Literature	GENERIF_SUMMARY	This study demonstrates the presence of MUC1 in psoriatic plaque and suggests a possible role for MUC1 during the motility, migration and survival of human keratinocytes
Literature	GENERIF_SUMMARY	High mucin expression is associated with mucinous adenocarcinoma in rectal carcinoma.
Literature	GENERIF_SUMMARY	Results show that MUC1 trans-membrane C-terminal subunit induces LIN28B expression, downregulates let-7 signaling and thereby promotes epithelial-mesenchymal transition and stemness in non-small cell lung cancer cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data indicate that mucin 1 (MUC1) forms a complex with NF-kappaB p65 transcription factor and binds to urokinase-type plasminogen activator (uPA) promoter, and thereby induces uPA expression.
Literature	GENERIF_SUMMARY	The specificity for IMP3, GLUT-1, and EMA for differentiating malignant mesotheliomas and benign mesothelial proliferations was 100%
Literature	GENERIF_SUMMARY	MUC1 and VEGF mRNA positivity in blood seem to be indicators of unfavorable response and poor PFS and OS in patients with advanced NSCLC treated with gefitinib.
Literature	GENERIF_SUMMARY	Serum Her2 level combined with CA-15-3 were strong and independent indicators for a worse disease free survival in patients with early breast cancer.
Literature	GENERIF_SUMMARY	Detection of glyco-mucin profiles improves specificity of MUC16 and MUC1 biomarkers in ovarian serous tumours.
Literature	GENERIF_SUMMARY	MUC1 rs4072037 allele G was protective against development of gastric cancer
Literature	GENERIF_SUMMARY	MUC1 oncogenic mucin alters proliferation, migration, and invasion properties of pancreatic cancer cells and that these effects are mediated by p42-44 MAPK, Akt, Bcl-2 and MMP13 pathways.
Literature	GENERIF_SUMMARY	MUC1 mediates autocrine TGF-beta signaling by activating the JNK/AP-1 pathway in hepatocellular carcinoma cells.
Literature	GENERIF_SUMMARY	NCOA3 upregulates mucin 1 and 4 via transcriptional and post-translational changes in pancreatic cancer.
Literature	GENERIF_SUMMARY	The combined detection of CYFRA21-1, CA125, SCC and CA153 is an effective way to distinguish benign and malignant SPN, and will find an important clinical application in the early diagnosis of SPN.
Literature	GENERIF_SUMMARY	Localization of MUC1 protein varied among breast cancer subtypes.
Literature	GENERIF_SUMMARY	MAMs MUC1 and MUC16 contribute to the maintenance of immune homeostasis at the ocular surface by limiting TLR-mediated innate immune responses.
Literature	GENERIF_SUMMARY	Letter
Literature	GENERIF_SUMMARY	Our data demonstrated that MUC1 and CD176 might be promising biomarkers for thyroid cancer
Literature	GENERIF_SUMMARY	KL-6 levels were higher and CC16 levels were lower in infants with poor neurodevelopmental outcome compared with those infants who had favourable neurodevelopmental outcome.
Literature	GENERIF_SUMMARY	There was no significant difference in MUC1 expression between chronic rhinosinusitis patients with and

Main Category	Category	Terms
		without bacterial biofilm formation.
Literature	GENERIF_SUMMARY	MUC1 rs4072037 was shown to decrease the risk of stomach cancer in a Han Chinese population.
Literature	GENERIF_SUMMARY	These results support a model in which MUC1-C-induced TAK1short right arrowNF-kappaB signaling contributes to intestinal inflammation and colon cancer progression.
Literature	GENERIF_SUMMARY	The present study indicates a novel mechanism by which MUC1 promotes cancer progression in human hepatocellular carcinoma
Literature	GENERIF_SUMMARY	Serum KL-6 levels were elevated in 8.1% of patients with rheumatoid arthritis treated with biological agents.
Literature	GENERIF_SUMMARY	Both blood and tissue measurements of MUC-1 and GdA were significantly lower in women with recurrent implantation failure than in fertile women during the implantation window.
Literature	GENERIF_SUMMARY	We show for the first time that MUC1/SEC and MUC1/Y are expressed in labial salivary glands of both Sjogren's syndrome patients and non-Sjogren's syndrome sicca subjects.
Literature	GENERIF_SUMMARY	Findings suggest that C1GALT1 enhances breast cancer malignant progression through promoting MUC1-C/beta-catenin signaling pathway.
Literature	GENERIF_SUMMARY	3Score1-MUC1 was profoundly expressed in the blood streams of patients with recurrent and/or metastatic breast cancer.
Literature	GENERIF_SUMMARY	The mean CA 15-3 serum level was significantly higher in the confirmed positive patients of PET/CT results than in the confirmed negative ones, while there were no significant differences in the serum levels of CEA and CA 125 of both groups.
Literature	GENERIF_SUMMARY	As engagement of MGL in the absence of TLR triggering may lead to anergy, the binding of MUC1-STn to MGL may be in part responsible for some of the characteristics of STn expressing tumours
Literature	GENERIF_SUMMARY	Flagellin drives NEU1 to desialylate MUC1, thereby increasing its adhesiveness for Pseudomonas aeruginosa and its shedding
Literature	GENERIF_SUMMARY	The MUC1 extracellular domain might be translocated to the cell nucleus in breast, head and neck, colon cancer, and benign breast disease.This domain protein core may be involved in transcriptional and posttranscriptional processes associated with stress.
Literature	GENERIF_SUMMARY	A pancreaticobiliary histomolecular classification (PB) was defined as a sample with pancreaticobiliary histology, positive MUC1 and negative CDX2 expression
Literature	GENERIF_SUMMARY	Data supports a novel regulation of COX-2 gene by MUC1 in pancreatic ductal adenocarcinoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	identified miR-29a and miR-330-5p as two new tumor suppressive miRNAs that inhibit the expression of MUC1 oncogenic mucin in pancreatic cancer cells.
Literature	GENERIF_SUMMARY	High Mucin 1 expression is associated with cutaneous T-cell lymphoma.
Literature	GENERIF_SUMMARY	Studies demonstrated that MUC1-mediated JNK activation not only enhances the phosphorylation of Smad2 C-terminal at Ser-465/467 site (Smad2C) through TGF-beta/TbetaRI, but also enhances the phosphorylation of Smad2 linker region at Ser-245/250/255 site.
Literature	GENERIF_SUMMARY	we showed that two serum magnesium associated loci, MUC1 and TRPM6, had significant effect modification with progestin use and insulin levels, respectively, in European Americans.
Literature	GENERIF_SUMMARY	All tumors expressed cytoplasmic and membranous mucin 1
Literature	GENERIF_SUMMARY	Previously reported association for rs4072037 in MUC1 with p=6.59x10(-8) for total Gastric cancer and similar estimates for cardia and non-cardia cancers was confirmed.
Literature	GENERIF_SUMMARY	MUC1-BSA-PD-NRs substantially decreased cell viability in photoirradiated MCF-7 cell lines vs. MUC1- MDA-MB-231 breast cancer cells
Literature	GENERIF_SUMMARY	Diagnostic efficacy of CA 15-3 and CEA in the early detection of metastatic breast cancer-A retrospective analysis
Literature	GENERIF_SUMMARY	Preoperative serum levels of CEA and CA15-3 are independent prognostic parameters for breast cancer.
Literature	GENERIF_SUMMARY	the diagnostic value of serum cancer antigen (CA)125, CA19-9 and CA15-3 concentrations in endometriosis, was evaluated.
Literature	GENERIF_SUMMARY	analysis of LGALS3, PHB2, MUC1, and GK2 expression with CA15-3 in early-stage breast cancer
Literature	GENERIF_SUMMARY	Genetic Variants of MUC1 Gene are not Associated with Colorectal Cancer.
Literature	GENERIF_SUMMARY	Data identified significantly lower expression of miR-1291 in esophageal squamous cell carcinoma (ESCC) and acts as a tumor suppressor by targeting MUC1.
Literature	GENERIF_SUMMARY	Among patients with elevated concentrations of the CA 15-3, CEA, and TPA concordance could be observed between serial changes of the hypermethylated RASSF1A gene and the protein markers
Literature	GENERIF_SUMMARY	suggest that the binding of galectin-3 to MUC1 plays a key role in MUC1-mediated signaling
Literature	GENERIF_SUMMARY	Immunostaining for CK7 and EMA can be used to

Main Category	Category	Terms
		differentiate interlobular bile ducts from ductular proliferation in patients with cholestasis.
Literature	GENERIF_SUMMARY	the pooled evidence suggested that MUC1 expression was significantly correlated with CRC metastasis.
Literature	GENERIF_SUMMARY	MUC1 gene silencing induces growth inhibition in SMMC7721 cells through Baxmediated mitochondrial and caspase-8-mediated death receptor apoptotic pathways.
Literature	GENERIF_SUMMARY	High WFA-Sialylated MUC1 is associated with Biliary Tract Cancer.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	Serum KL-6 was not significantly different between patients with connective tissue disease-interstitial lung disease, chronic fibrosing interstitial pneumonia patients and healthy controls.
Literature	GENERIF_SUMMARY	A simple diagnostic index including epithelial membrane antigen and fibronectin levels can be used for early detection of hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	immunoexpression of different epitopes (external and cytoplasmic) of MUC1 expression are altered in tubes with ectopic pregnancy compared with normal tubes, suggesting an association to explain its etiology
Literature	GENERIF_SUMMARY	The KL-6 concentration of pancreatic juice may be as useful as PJC for diagnosing PDACs.
Literature	GENERIF_SUMMARY	this study demonstrates that first-trimester spontaneous abortion is associated with increases in MUC1 expression in villi and decreases in the decidual tissues
Literature	GENERIF_SUMMARY	silencing muc-1 reduced squamous cell carcinoma cellular colony forming ability, migration and invasion
Literature	GENERIF_SUMMARY	High level KL-6 in infected bronchoalveolar lavage fluid samples may reflect the injury of alveolar epithelium caused by M. pneumoniae.
Literature	GENERIF_SUMMARY	There were differences in MUC1 and c-myc expression in benign and papillary thyroid carcinoma.
Literature	GENERIF_SUMMARY	These results indicate that in response to Pseudomonas aeruginosa or flagellin, EGFR associates with and tyrosine phosphorylates MUC1-CT in primary normal human bronchial epithelial cells, leading to increased MUC1-CT association with TLR5
Literature	GENERIF_SUMMARY	Our findings suggested that MUC1 plays an important role in CCL21-CCR7-induced lymphatic metastasis and may serve as a therapeutic target in esophageal squamous cell carcinoma .
Literature	GENERIF_SUMMARY	Data indicate the potential of functionalized Apt-carbon nanotubes conjugates for increasing the induction of apoptosis in Mucin-1 (MUC1) positive tumor cells by

Main Category	Category	Terms
		suppression of Bcl-xL transcript.
Literature	GENERIF_SUMMARY	he present meta-analysis indicated that EMA may be a useful diagnostic tool with good sensitivity and specificity for differentiating malignant effusions from benign effusions.
Literature	GENERIF_SUMMARY	In the comparison of all the groups in the study, significantly higher levels of KL-6 were determined in the CTD-related interstitial lung diseases group than in either the CTD without pulmonary involvement group or the healthy control group. The level of KL-6 as a predictive factor could be used to identify the clinical development of interstitial lung disease before it is detected on imaging modality
Literature	GENERIF_SUMMARY	our data suggest a novel molecular mechanism by which tMUC1 may modulate NRP1-dependent VEGFR signaling in PDA cells.
Literature	GENERIF_SUMMARY	Using differential pulse voltammetry (DPV) technique, CA 15-3 can be selectively detected as low as 0.01 +/- 0.07 U/mL in human serum samples. Additionally, the proposed CA 15-3 immunosensor showed excellent selectivity and better stability in human serum samples, which demonstrated that the proposed immunosensor has potentials in proteomic researches and diagnostics
Literature	GENERIF_SUMMARY	MUC1 expression significantly correlated with that of MYC and its target genes in human KRAS mutant NSCLC tumors.
Literature	GENERIF_SUMMARY	In saliva the correlation between autoantibodies anti-MUC1 and anti-HER2 more important than in serum.
Literature	GENERIF_SUMMARY	In this phase I/II study, 14 high-risk disease-free ovarian (OC) and breast cancer (BC) patients after completion of standard therapies were vaccinated with MUC1, ErbB2 and carcinoembryonic antigen (CEA) HLA-A2+-restricted peptides and Montanide.
Literature	GENERIF_SUMMARY	Analysis of microarray data sets further demonstrated that MUC1 levels positively correlate with MYC expression in multiple myeloma progression
Literature	GENERIF_SUMMARY	results suggest that the MUC1 rs4072037G allele may be a low-penetrating protection factor for gastric cancer risk in Chinese populations
Literature	GENERIF_SUMMARY	MUC1-C binds directly with CD44v and in turn promotes stability of xCT in the cell membrane
Literature	GENERIF_SUMMARY	identified this type of MUC1 mutation in 9.5 % of families from an ADTKD Italian cohort; larger studies are needed to better define the criteria for genetic testing for this type of mutation
Literature	GENERIF_SUMMARY	that, in adenocarcinomas and borderline ovarian tumors, the major carriers of SLe(a) and SLe(x) are MUC16 and/or MUC1 (100 and 92 % for SLe(a)
Literature	GENERIF_SUMMARY	Only EMA was significantly associated with the

Main Category	Category	Terms
		expressions in circulating tumor cells (CTCs) and tissue. CTC detection was associated with higher T stage and portal vein invasion in hepatocellular carcinomas patients
Literature	GENERIF_SUMMARY	MUC1-driven EGFR expression and signaling regulates proliferation of endometrial cancer cells.
Literature	GENERIF_SUMMARY	MUC1 rs4072037 A/A genotype is associated with more severe pulmonary dysfunction and a higher rate of disease progression in PAP patients.
Literature	GENERIF_SUMMARY	Validation results indicate an accurate and reliable test for clinically establishing the molecular diagnosis of mucin-1 kidney disease with 100% sensitivity and specificity across 275 tests called.
Literature	GENERIF_SUMMARY	Overexpression of MUC1 is associated with Breast Cancer Micrometastasis.
Literature	GENERIF_SUMMARY	High MUC1 in either primary tumor or lymph node metastases correlates significantly with unfavorable tumor features and survival. Overexpression of MUC1 in the metastases of a subset of prostate cancer patients may have therapeutic potential.
Literature	GENERIF_SUMMARY	Studies showed that MUC1 is a prognostic factor in gastric cancer, which acts as a marker of poor outcome in patients with gastric cancer.
Literature	GENERIF_SUMMARY	MUC1-C is of functional importance to induction of DNMT1 and DNMT3b and, in turn, changes in DNA methylation patterns in cancer cells.
Literature	GENERIF_SUMMARY	MUC1-C is upregulated in triple-negative breast cancer cells resistant to ABT-737 or ABT-263.
Literature	GENERIF_SUMMARY	MUC1 enhancement of ERK activation influences FRA-1 activity to modulate tumor migration, invasion and metastasis in a subset of pancreatic cancer cases
Literature	GENERIF_SUMMARY	MUC1-C activates the NF-kappaB p65 pathway, promotes occupancy of the MUC1-C/NF-kappaB complex on the DNMT1 promoter and drives DNMT1 transcription
Literature	GENERIF_SUMMARY	MUC1 plays an important role in Tumor-associated macrophage-induced lung cancer stem cell progression; pterostilbene may have therapeutic potential for modulating the unfavorable effects of TAMs in lung cancer progression
Literature	GENERIF_SUMMARY	MUC1 and MUC4 expression are increased by hypoxia and DNA hypomethylation; this status is statistically associated with development of distant metastasis, tumor stage and overall survival for pancreatic ductal adenocarcinoma (stage IIA and IIB) patients
Literature	GENERIF_SUMMARY	Elevated levels of KL-6 are Associated with Idiopathic Pulmonary Fibrosis.
Literature	GENERIF_SUMMARY	MUC1 was expressed in 238 (64 %) tumors and had no

Main Category	Category	Terms
		impact on outcome
Literature	GENERIF_SUMMARY	Differences in the apparent expression levels of CEA, CA153 and CA125 in patients with nipple discharge and healthy persons were validated.
Literature	GENERIF_SUMMARY	We have identified novel MUC1 class I peptides that bind to HLA-A*0201 molecules with significantly higher affinity and function than the native MUC1 peptides. These peptides elicited CTLs from normal donors, as well as breast cancer patients, which were highly effective in killing MUC1-expressing MCF-7 breast cancer cells.
Literature	GENERIF_SUMMARY	this study shows that basaloid squamous cell carcinoma and basal cell carcinoma of the head and neck can be readily distinguished by a limited panel consisting primarily of EMA, and supported by SOX2 and p16
Literature	GENERIF_SUMMARY	High Muc1 expression is associated with lung adenocarcinoma.
Literature	GENERIF_SUMMARY	The natural course of systemic sclerosis with interstitial lung disease is highly variable. Baseline serum KL-6 is a biomarker potentially useful for predicting FVC decline.
Literature	GENERIF_SUMMARY	Within this review, we highlight both the processes involved in the expression of aberrant glycan structures on mucins, as well as the potential downstream impacts on cellular signaling.
Literature	GENERIF_SUMMARY	These data suggest that imbalance of IL-22/IL-22R1 signaling regulates the pathogenesis of chronic rhinosinusitis with nasal polyps, including local eosinophilia, via alteration of MUC1 expression.
Literature	GENERIF_SUMMARY	This review aims to encompass a detailed characterization of MUC1 role in carcinogenesis, highlighting recent findings in cell differentiation and uncovering new evidences of MUC1 isoforms involvement in malignant phenotype. [Review]
Literature	GENERIF_SUMMARY	Data show that triple-negative breast cancer (TNBC) patients with pre-therapeutic serum high levels of both carcinoembryonic antigen (CEA) and cancer antigen 15-3 (CA15-3) had shorter overall survival (OS) and disease-free survival (DFS) rates than those in the low-level groups.
Literature	GENERIF_SUMMARY	This work defines a critical role for aberrantly glycosylated MUC1 and identifies an activating pathway that follows engagement of Siglec-9.
Literature	GENERIF_SUMMARY	The data demonstrated variation in corneal epithelial membrane-associated mucin expression between species, with the rabbit having a distinct expression pattern. These differences may be reflective of the environment, pathogen exposure or tear film dynamics of the respective species.
Literature	GENERIF_SUMMARY	it seems reasonable to further evaluate the prognostic

Main Category	Category	Terms
		value of circulating tumor cells and CA27.29 as a combined prognostic test of two potentially independent markers that might provide complementary prognostic information.
Literature	GENERIF_SUMMARY	Data suggest that GlcNAc6ST-2 (CHST4) is critical for biosynthesis of MECA-79 sulfated glycans in apical membranes of small-sized intrahepatic bile ducts and of cholangiolocellular carcinoma (CoCC) cells; MUC1 protein decorated with GlcNAc6ST-2-dependent MECA-79 sulfated glycans may serve as a useful CoCC marker.
Literature	GENERIF_SUMMARY	Results that overexpression and aberrant glycosylation of MUC1 on tumor cells influences many critical steps in tumor initiation, progression, and metastasis as well as in tumor immunosurveillance. This hypoglycosylated tumor form of MUC1 is also overexpressed during chronic inflammation and in premalignant lesions that progress to tumors. [review]
Literature	GENERIF_SUMMARY	The membrane-anchored MUC1 is expressed in the superficial layer of the oral mucosal epithelium, especially on the upper surface of epithelial cells. MUCI may be the anchoring protein of the salivary pellicle stabilization.
Literature	GENERIF_SUMMARY	Report MUC1 gene amplification in association with prostate cancer metastasis and the development of castration resistant prostate cancer.
Literature	GENERIF_SUMMARY	MUC1 protein expression was associated with adverse pathological features, although it was not an independent predictor of outcome after radical prostatectomy.
Literature	GENERIF_SUMMARY	These findings uncover a previously unrecognized role for MUC1-C in driving BMI1 expression and in directly interacting with this stem cell factor, linking MUC1-C with function of the PRC1 in epigenetic gene silencing.
Literature	GENERIF_SUMMARY	Data show that added synthetic long peptides derived from widely expressed oncoproteins MUC1 and HER2/neu were reliably presented to CD4+ T-cells and cross-presented to CD8+ T-cells.
Literature	GENERIF_SUMMARY	we evaluated the clinical performance of LOCItrade mark-based tumor marker assays CEA, CA19-9, CA15-3, CA125 and AFP in patients with gastrointestinal cancer and demonstrated their high diagnostic power
Literature	GENERIF_SUMMARY	MUC1 promotes hepatocellular carcinoma progression and tumorigenesis and mediates autocrine TGF-b signaling by activating the JNK/AP-1 pathway in hepatocellular carcinoma cells, and makes MUC1 and JNK as attractive targets for hepatocellular carcinoma therapy.
Literature	GENERIF_SUMMARY	Low MUC1 expression is associated with tongue squamous cell carcinoma.
Literature	GENERIF_SUMMARY	In this Molecular Pathways article, we briefly discuss

Main Category	Category	Terms
		the potential role of mucin synthesis in cancers, ways to improve drug delivery and disrupt mucin mesh to overcome chemoresistance by targeting mucin synthesis, and the unique opportunity to target the GCNT3 pathway for the prevention and treatment of cancers.
Literature	GENERIF_SUMMARY	CD44, ALDH1 and Twist1 were significantly overexpressed in involved ALN. The serum levels of CA153 in those patients were clearly increased and the survival rates decreased, which suggested that a combination of Twist1 in ALN and CA153 may function as an indicator for the prognosis of patients with breast cancer.
Literature	GENERIF_SUMMARY	The MUC1 oncoprotein was identified as the critical driver of (extracellular vesicle) EV-mediated myeloid-derived suppressor cells expansion in AML. microRNA miR34a acts as the regulatory mechanism by which MUC1 drives c-myc expression in AML cells and EVs.
Literature	GENERIF_SUMMARY	A previously unrecognized central role for MUC1-C in integrating PD-L1 activation with suppression of immune effectors and poor clinical outcome in non-small-cell lung cancer.
Literature	GENERIF_SUMMARY	These results revealed that serum WFA-sialylated MUC1 was associated with histological features of hepatocellular carcinoma and recurrence after curative therapy.
Literature	GENERIF_SUMMARY	this study implicates the MUC1 as a critical and dynamic component of the innate host response that limits the severity of influenza and provides the foundation for exploration of MUC1 in resolving inflammatory
Literature	GENERIF_SUMMARY	these data showed that sustained abnormal MUC1 induction accompanies failing epithelial repair, chronic inflammation and kidney fibrosis. In conclusion, MUC1 exerts opposite effects during kidney response to IR
Literature	GENERIF_SUMMARY	Findings provide evidence that rs4072037 MUC1 does not have a significant impact on the pathogenesis or prognosis of lung adenocarcinoma (ADC), whereas KL-6 serum levels, which might reflecting the molecular length of MUC1, are significantly associated with lung ADC.
Literature	GENERIF_SUMMARY	The presence of the MUC1 molecules containing TR subdomain (MUC1-TR) on the surface of low-invasive cancer cells leads to the increase in their transendothelial migration potency, while the addition of the IR subdomain to the MUC1-TR molecule (MUC1-IR-TR) restores their natural low invasiveness.
Literature	GENERIF_SUMMARY	Studied predictive use of mucin 1 (KL-6) serum level as a biomarker in development of bronchopulmonary dysplasia in preterm infants.
Literature	GENERIF_SUMMARY	In stage IV breast cancer, circulating antiMUC1 antibody was found to bind serum MUC1 antigen, although their

Main Category	Category	Terms
		compatibility was low. No significant difference was found in the affinity of the antiMUC1 antibody between stage IV breast cancer and earlystage breast cancer.
Literature	GENERIF_SUMMARY	Results suggest that MUC1 serves as a metabolic regulator in triple-negative breast cancer, facilitating the metabolic reprogramming of glutamine utilization that influences triple-negative breast cancer tumor growth.
Literature	GENERIF_SUMMARY	MUC1 gene interference was done to A549 cells to show its role in sensitivity of lung cancer cells to TNFalpha and DEX. Results of our experiments indicate that MUC1 may regulate the influence of inflammatory mediators in effects of glucocorticoids (GCs), as a regulatory target to improve therapeutics.
Literature	GENERIF_SUMMARY	Mucin 1 is present in intervertebral disc tissue, and its expression is altered in disc degeneration.
Literature	GENERIF_SUMMARY	MUC1 expression correlates with tumor metastatic recurrence in postoperative esophageal squamous cell cancer patients
Literature	GENERIF_SUMMARY	In conclusion, this meta-analysis suggested that rs4245739 polymorphism in the MUC1 gene may play a pivotal role in the pathogenesis of GC, especially for white populations
Literature	GENERIF_SUMMARY	Studied and analyzed MUC1 in specimens from patients with cholangiocarcinoma and pancreatic ductal adenocarcinoma. Using lectin microarray analysis, analyzed and found different glycoforms of MUC1 could distinguish between different cancers and degree of malignancy.
Literature	GENERIF_SUMMARY	In malignant epithelial ovarian tumors, the positive expression rates of Lewis(y) antigen and MUC1 were 88.33 and 86.67%, respectively, which were markedly higher than those in borderline (60.00 and 53.33%, P
Literature	GENERIF_SUMMARY	findings show that transmembrane mucins are receptors for the aggregative adherence fimbriae (AAF) adhesins of enteroaggregative Escherichia coli on the intestinal epithelium; demonstrate that the AAFs elicit intestinal inflammation through MUC1-mediated host cell signaling
Literature	GENERIF_SUMMARY	MUC1 is aberrantly overexpressed on the entire tumor cell surface of most human cancers; MUC1 induces tamoxifen resistance in estrogen receptor-positive breast cancer
Literature	GENERIF_SUMMARY	As most markers were also elevated in benign breast diseases, only cancer antigen 15-3 showed significant differences to cancer patients (p
Literature	GENERIF_SUMMARY	Detection of MUC1 and HS3ST2 promoter methylation status appears to be useful molecular markers for assessing the progressive state of the disease and could be helpful in discriminating breast cancer molecular subtypes. These results validate the methylation-based microarray analysis, thus trust their

Main Category	Category	Terms
		output in the future.
Literature	GENERIF_SUMMARY	MUC1-mediated nucleotide metabolism plays a key role in facilitating radiation resistance in pancreatic cancer and targeted effectively through glycolytic inhibition
Literature	GENERIF_SUMMARY	These findings indicate that decitabine intensifies MUC1-C inhibition induced redox imbalance and provides a novel combination of targeted and epigenetic agents for patients with Cutaneous T-cell lymphoma
Literature	GENERIF_SUMMARY	As MUC1 and galectin-3 are both commonly overexpressed in most types of epithelial cancers, their interaction and impact on EGFR activation likely makes important contribution to EGFR-associated tumorigenesis and cancer progression.
Literature	GENERIF_SUMMARY	In this paper, a dual-target electrochemical aptasensor has been developed for simultaneous detection of carcinoembryonic antigen and mucin-1 based on metal ion electrochemical labels and Ru(NH3)6(3+) electronic wires
Literature	GENERIF_SUMMARY	Findings suggest a mechanism linking mucin-type core 3 O-glycan to the mesenchymal-epithelial transition (MET) of colorectal cancer (CRC) cells via MUC1/p53/miR-200c-dependent signaling cascade.
Literature	GENERIF_SUMMARY	Studies demonstrate that MUC1-C-terminal beta subunit drives EZH2 expression in TNBC, NSCLC and other types of carcinoma cells. MUC1-C activates the EZH2 promoter through induction of the pRB-->E2F pathway. MUC1-C binds directly to the EZH2 CXC region adjacent to the catalytic SET domain and associates with EZH2 on the CDH1 and BRCA1 promoters.
Literature	GENERIF_SUMMARY	the observed G1 phase arrest completely agrees with the metabolomics results; MUC1-overexpressing cells under glucose limitation have an altered glutamine metabolism that results in a disruption in de novo pyrimidine synthesis that negatively impacts DNA replication. Moreover, our results provide a clear explanation for the observed glucose dependency of MUC1-overexpressing cells.
Literature	GENERIF_SUMMARY	Pseudomonas aeruginosa - stimulated MUC1 expression in THP-1 macrophages is regulated mainly through the TLR4-p38 signaling pathway.
Literature	GENERIF_SUMMARY	silencing MUC1 expression inhibited migration and invasion, and induced apoptosis of PANC-1 cells via downregulation of Slug and upregulation of Slug dependent PUMA and E-cadherin expression.
Literature	GENERIF_SUMMARY	In uninflamed CD ileum and IBD colon, most barrier gene levels restored to normal, except for MUC1 and MUC4 that remained persistently increased compared with controls. Genetic and transcriptomic dysregulations of key epithelial barrier genes and components in IBD. In particular MUC1 and MUC4, play an essential role in the pathogenesis of IBD and could

Main Category	Category	Terms
		represent interesting targets for treatment.
Literature	GENERIF_SUMMARY	Results identified MUC1 as a novel target of 14-3-3zeta in lung adenocarcinoma. Its high expression is associated with poor survival in lung adenocarcinoma patients.
Literature	GENERIF_SUMMARY	Data suggest that ositive Mucin-1 (MUC1) expression in cell block cytology specimens may be associated with progressive dilation of the main and ectatic branches of pancreatic ducts.
Literature	GENERIF_SUMMARY	MUC1 role in pulmonary fibrosis
Literature	GENERIF_SUMMARY	Correlation was also observed in the % change of CA 15-3 and CA 27.29 results between consecutive specimens for individual patients. Using doubling or halving thresholds (i.e., 100% increase or 50% decrease), concordance in % change was observed between CA 15-3 and CA 27.29 in approximately 90% of cases. Individual patient results trended similarly across both markers over time
Literature	GENERIF_SUMMARY	MUC1 up-regulation is associated with castration-resistant prostate cancer and bone metastasis.
Literature	GENERIF_SUMMARY	this paper shows the role of IgG and Fcgamma receptor genes in endogenous antibody responses to mucin 1 in a large multiethnic cohort of Brazilian patients with breast cancer
Literature	GENERIF_SUMMARY	Frameshift mutation in MUC1 is associated with autosomal dominant tubulointerstitial kidney disease.
Literature	GENERIF_SUMMARY	MUC1 contributes to immune escape in an aggressive form of triple-negative breast cancer.MUC1 drives PD-L1 expression in triple-negative breast cancer cells.
Literature	GENERIF_SUMMARY	the proposed ECL immunosensor opened a new era for sensitive CA15-3 evaluation and offered a promising platform for clinical breast cancer diagnostics.
Literature	GENERIF_SUMMARY	findings suggest that these pulmonary markers could be useful to assess CAP severity and, especially YKL-40 and CCL18 by helping predict CAP caused by atypical pathogens
Literature	GENERIF_SUMMARY	Threonyl-tRNA synthetase regulates MUC1 biosynthesis in pancreatic cancer cells.MUC1 role in the pancreatic cancer cell migration.
Literature	GENERIF_SUMMARY	Quercetin suppressed breast cancer stem cell proliferation, self-renewal, and invasiveness. It also lowered the expression levels of proteins related to tumorigenesis and cancer progression, such as aldehyde dehydrogenase 1A1, C-X-C chemokine receptor type 4, mucin 1, and epithelial cell adhesion molecules.
Literature	GENERIF_SUMMARY	MUC1-C thereby contributes to the integration of PRC2 and PRC1-mediated repression of tumor suppressor genes, such as CDH1, CDKN2A, PTEN and BRCA1. Like

Main Category	Category	Terms
		PRC2 and PRC1, MUC1-C is associated with the epithelial-mesenchymal transition (EMT) program, cancer stem cell (CSC) state, and acquisition of anticancer drug resistance. [review]
Literature	GENERIF_SUMMARY	The CA 15-3 assay using 115D8-DF3 antibody pair is more suitable for monitoring therapy in pregnancy-associated breast cancer
Literature	GENERIF_SUMMARY	We have analysed the tumour-associated carbohydrate antigens sialyl-Lewis x (SLe(x)) and sialyl-Tn (STn) on MUC1 and MUC5AC in Pancreatic adenocarcinoma (PDAC)tissues. immunoprecipitation of MUC5AC from positive PDAC tissues and subsequent SLe(x) immunodetection confirmed the presence of SLe(x) on MUC5AC. Altogether, MUC5AC-SLe(x) glycoform is present in PDAC and can be regarded as potential biomarker.
Literature	GENERIF_SUMMARY	High MUC1 expression is associated with breast cancer metastasis.
Literature	GENERIF_SUMMARY	In the in vitro tests, JFD-WS effectively inhibited HUVEC proliferation, migration, tube formation and VEGFR2 phosphorylation. Additionally, JFD-WS inhibited the formation of blood vessels in chick chorioallantoic membrane. While inhibiting the xenograft tumor growth in experimental mice, JFD-WS decreased the plasma MUC1 levels
Literature	GENERIF_SUMMARY	Mutated MUC1 (insC) affects vesicular transport in renal epithelial cells. It changes gene expression of vesicular transport-associated proteins, for example VTA1.
Literature	GENERIF_SUMMARY	The suprachoroidal region of the eye comprises vascular channels, melanocytes, and thin fibroblasts with elongated cytoplasm that are positioned directly adjacent to the densely collagenous sclera. Granular, diffuse, and cytoplasmic EMA expression was seen in suprachoroidal fibroblasts, but this was not contiguous with the similar pattern of EMA expression in adjacent perineurium and arachnoid.
Literature	GENERIF_SUMMARY	KL-6 is an accurate biomarker for the diagnosis of interstitial lung disease in systemic sclerosis.
Literature	GENERIF_SUMMARY	This study demonstrates that precise positioning of the causative mutation(s) and identification of other coding and noncoding sequence variants in autosomal dominant tubulointerstitial kidney -MUC1 is feasible. SMRT sequencing could provide a powerful tool to uncover potential factors encoded within the VNTR that associate with intra- and interfamilial phenotype variability of MUC1 related kidney disease.
Literature	GENERIF_SUMMARY	we wanted to explore whether STAT3 can be related to lymph node micrometastasis of non-small cell lung cancer (NSCLC). To address this question, we evaluated the expression of MUC1 mRNA in the lymph node samples of NSCLC to determine micrometastasis.

Main Category	Category	Terms
		Then, we evaluated what role STAT3 overexpression plays in lymph node micrometastasis of NSCLC.
Literature	GENERIF_SUMMARY	Interstitial lung disease patients with clubbing had lower oxygen levels, higher serum KL-6 levels, and lower pulmonary function than those without clubbing.
Literature	GENERIF_SUMMARY	MUC1 was a potential molecular target may help explain the role of lincRNA-ROR/miR-145 for invasion and metastasis in Triple-negative breast cancer cell lines.
Literature	GENERIF_SUMMARY	Through IL-11-STAT3/ERK signaling by upregulating MUC1.
Literature	GENERIF_SUMMARY	A colon score derived from serum CEA, CA19-9, CK1 and MUC1 is a potential valuable non-invasive index that could be used for detection and screening early stage colon cancer patients
Literature	GENERIF_SUMMARY	The glycosylation level of CA153 was found to increase with increasing breast cancer stage in the sandwich assay. The assay system appeared to efficiently discriminate breast cancer stage I (sensitivity
Literature	GENERIF_SUMMARY	Mucin Expression in Colorectal Cancer (CRC)
Literature	GENERIF_SUMMARY	The main genetic cause of autosomal dominant tubulointerstitial kidney disease in a Spanish cohort is a MUC1 pathogenic mutation.
Literature	GENERIF_SUMMARY	Results show MUC1 expression highly expressed at mRNA and protein levels in esophageal squamous cell carcinoma (ESCC). MUC1 expression correlated with tumor invasion, lymph node metastasis, and TNM staging.
Literature	GENERIF_SUMMARY	MUC-1, CK7, and c-p27 immunostaining can be used as the predictive markers for esophageal cancer progression from Barrett's esophagus.
Literature	GENERIF_SUMMARY	Decreased expression of MUC1 is an independent marker for endometrial receptivity in recurrent implantation failure.
Literature	GENERIF_SUMMARY	KL-6 but not CCL-18 is predictive of early systemic sclerosis -related interstitial lung disease progression.
Literature	GENERIF_SUMMARY	Mucin 1-CT expression was downregulated in bronchial epithelial cells and peripheral blood neutrophils from severe asthma patients compared with mild asthma and healthy subjects. Mucin 1-CT modulates corticosteroid efficacy in vitro and in vivo asthma models.
Literature	GENERIF_SUMMARY	The expression profile of studied Mucins MUC16 and MUC1 and truncated O-glycans was not associated with the site of origin of ovarian cancer (OVCA) cell lines
Literature	GENERIF_SUMMARY	High MUC1 expression is associated with cervical cancer.

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Literature	GENERIF_SUMMARY	serum level not related to breast cancer subgroup
Literature	GENERIF_SUMMARY	MiR-326 functioned as a tumor suppressor in PCa by negatively regulating MUC1.
Literature	GENERIF_SUMMARY	From an increase in the number of tandem repeats through the decrease of sialylation to the downregulation of MUC1 expression level
Literature	GENERIF_SUMMARY	MUC1 was intensely positive in 46% and negative in 37.9% of colorectal tumors. Loss of MUC1 expression was more frequent in cases with disease recurrence or death, as compared with patients with stable disease, in whom intense positivity was more frequently seen. These findings disagree with the majority of previous studies.
Literature	GENERIF_SUMMARY	A combination of tissue overexpression of MUC1, reduced MUC2 expression, and high ratio of MUC1/MUC2 is a factor of poor prognosis in colorectal cancer (CRC) patients. MUC2 tissue expression allows to differentiate mucinous and non-mucinous CRC subtypes.
Literature	GENERIF_SUMMARY	To achieve this, using semi-quantitative mass spectrometry, we found MUC1 to be significantly and durably upregulated in response to erlotinib, an EGFR-targeting treatment. MUC1 upregulation was regulated transcriptionally, involving PI3K-signaling and STAT3
Literature	GENERIF_SUMMARY	this study shows that changes in serum KL-6 levels are associated with the development of chronic lung allograft dysfunction in lung transplant recipients
Literature	GENERIF_SUMMARY	MUC1 expression is involved in accelerated growth of EMM1 cells.
Literature	GENERIF_SUMMARY	MUC1-CT (cytoplasmic tail) expression was downregulated in lung tissue, bronchial epithelial cells and lung neutrophils from smokers and in COPD and it was correlated with their FEV1%
Literature	GENERIF_SUMMARY	Study demonstrated that MCU1 is frequently overexpressed in breast cancer and abnormally high expression of MUC1 indicates poor prognosis. Subsequent data mining across multiple large databases demonstrated a positive association between MUC1 mRNA expression CREB3L4 in breast cancer tissues. Results indicated that MUC1 transcript expression may regulate tumor invasion and metastasis associated with CREB3L4 transcription.
Literature	GENERIF_SUMMARY	Findings indicate that linc01296/miR-26a/GALNT3 axis involves in the progression of CRC cells, illuminating the possible mechanism mediated by O-glycosylated MUC1 via PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Baseline elevated serum KL-6 may predict deterioration of lung function of systemic sclerosis-interstitial lung disease patients with mild lung injury.
Literature	GENERIF_SUMMARY	The most strong relationship was registered between

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		levels of expression of genes MUC I and IL32 (r-0.72; p=0.0001). In comparison with MUC I negative breast carcinoma, MUC I positive breast carcinoma characterized by higher level of mRNA of pro-inflammatory cytokines IL32 and capital IE, Cyrilliccapital TE, Cyrilliccapital A, Cyrillic
Literature	GENERIF_SUMMARY	Serum KL-6 is a useful biomarker for assessing the disease activity of myositis-associated interstitial lung disease.
Literature	GENERIF_SUMMARY	These results suggest that certain proteins, such as MUC1, are selectively enriched in the exosome compartment. The mechanisms for their preferential localization and their biological roles remain to be studied
Literature	GENERIF_SUMMARY	Study demonstrates for the first time that MUC1 augments the underlying JAK/STAT signaling, enhances IFITM1 expression, and promotes an aggressive phenotype in aromatase inhibitor (AI)-resistant breast cancer cells. Clinically, coexpression of MUC1 and IFITM1 positively correlated with aggressive breast cancer and resistance to therapy.
Literature	GENERIF_SUMMARY	Calgranulin B and KL-6 in BAL proved to be reliable biomarkers of diopathic pulmonary fibrosis (IPF) and non-specific interstitial pneumonia (NSIP) and to have prognostic meaning, discriminating severe and advanced patients. The combination of the two biomarkers can facilitate the stratification of severity.
Literature	GENERIF_SUMMARY	The positive expression of MUC1 in human duodenal cancer cell line HuTu80 was confirmed by RT-PCR and flow cytometry. The outcome of duodenal adenocarcinoma patients with high MUC1 expression was significantly worse than those without MUC1 expression.
Literature	GENERIF_SUMMARY	Serum krebs von den Lungen-6 ()levels were increased in connective tissue disease-associated interstitial lung disease (CTD-ILD) patients.
Literature	GENERIF_SUMMARY	BRG1 activates MUC1 transcription by interacting with STAT1 and RelA.
Literature	GENERIF_SUMMARY	Serum KL-6 is associated with the severity of interstitial lung disease in Chinese patients with polymyositis and dermatomyositis
Literature	GENERIF_SUMMARY	patients suffering lung, breast, ovarian cancers, nephrotic syndrome, type 2 diabetes, endometrial cancer, coronary heart disease, cervical cancer, uremia, and other 12 diseases plus healthy controls >65 years old had significantly (p1.30) increased median serum CA153 levels compared to that of healthy controls.
Literature	GENERIF_SUMMARY	MUC1 is related to the occurrence and development of salivary gland acinic cell carcinoma
Literature	GENERIF_SUMMARY	A biosensor capable of identifying low quantities of breast cancer cells by electrical impedance

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		spectroscopy.
Literature	GENERIF_SUMMARY	LINC00909 could act as a ceRNA to interact with miR-194 and thereby up-regulate the expression of MUC1-C.
Literature	GENERIF_SUMMARY	Binding of galectin-3 to MUC1 enhanced Sp1 recruitment, leading to Trop-2 induction.
Literature	GENERIF_SUMMARY	The length of MUC1 can influence IL-1beta production, thus providing the first evidence of an immune-modulatory role of MUC1 VNTR polymorphisms in human macrophages.
Literature	GENERIF_SUMMARY	Diagnostic value of CA-153 and CYFRA 21-1 in predicting intraocular metastasis in patients with metastatic lung cancer.
Literature	GENERIF_SUMMARY	Up-regulation of miR-136 reduced the survival rate, suppressed colony formation ability and induced apoptosis of esophageal squamous cell carcinoma (ESCC) cells under irradiation, which was reversed by upregulated mucin 1 (MUC1).
Literature	GENERIF_SUMMARY	MUC1 downregulates the expression of GALNT5 in pancreatic cancer by modifying the promoter occupancy of transcription factors through its cytoplasmic domain.
Literature	GENERIF_SUMMARY	Overexpression of KL6 is associated with clinically amyopathic dermatomyositis and interstitial lung disease.
Literature	GENERIF_SUMMARY	These findings uncover a master role for MUC1-C in driving the induction of TWIST1.
Literature	GENERIF_SUMMARY	MUC-1 aptamer targeted superparamagnetic iron oxide nanoparticles for magnetic resonance imaging of pancreatic cancer in vivo and in vitro experiment.
Literature	GENERIF_SUMMARY	upregulation of MUC1 in involved psoriatic lesion compared to uninvolved and normal skin may suggest MUC1 role in pathogenesis of psoriasis especially early stages; MUC1 may be responsible for less severity of psoriasis in old aged patients
Literature	GENERIF_SUMMARY	Among individuals since serine- and threonine-rich peptide sequences (STPS) in the cancer-specific MUC-1 antigen are surrogate ligands for this antibody.
Literature	GENERIF_SUMMARY	MiR210HG sponges miR-1226-3p to facilitate invasive breast cancer cell invasion and metastasis by regulating mucin-1c and EMT pathway
Literature	GENERIF_SUMMARY	These findings define a previously unrecognized role for MUC1-C in suppression of RASSF1A in cancer
Literature	GENERIF_SUMMARY	The potential prognostic relevance of EP3 expression for OS in TA-MUC1 negative patients might reflect an interplay between the COX and the MUC1 pathway, as it has been shown that MUC1 could induce COX2 expression.

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Literature	GENERIF_SUMMARY	the level of serum KL-6 in the silicosis group was significantly decreased compared with the control group, while the level of KL-6 in the serum of the silica dust-exposed group was significantly increased
Literature	GENERIF_SUMMARY	MUC1-C activates the NuRD complex to drive dedifferentiation.
Literature	GENERIF_SUMMARY	The specific expression of Mucin 1 (MUC1) in the ciliated cells in receptive control women is higher than that of women with reproductive failure during implantation window.
Literature	GENERIF_SUMMARY	Continuing low-volume ventilation during cardiopulmonary bypass could decrease the expression level of KL-6, while it has no significant influence on perioperative outcomes.
Literature	GENERIF_SUMMARY	Separation and detection of minimal length glycopeptide neoantigen epitopes centering the GSTA region of MUC1 by liquid chromatography/mass spectrometry.
Literature	GENERIF_SUMMARY	serum KL-6 is an important biomarker for the diagnosis of interstitial lung disease associated with connective tissue diseases (CTD-ILD) and when the level of KL-6 is increased, the ILD should be alert
Literature	GENERIF_SUMMARY	Role of glycosylation on the ensemble of conformations in the MUC1 immunodominant epitope.
Literature	GENERIF_SUMMARY	Serum KL-6 level as a biomarker of interstitial lung disease in childhood connective tissue diseases
Literature	GENERIF_SUMMARY	MUC1 as a target for CAR-T therapy in head and neck squamous cell carinoma.
Literature	GENERIF_SUMMARY	Cross-talk between Colon Cells and Macrophages Increases ST6GALNAC1 and MUC1-sTn Expression in Ulcerative Colitis and Colitis-Associated Colon Cancer.
Literature	GENERIF_SUMMARY	MUC-1 recognition-based activated drug nanoplatform improves doxorubicin chemotherapy in breast cancer.
Literature	GENERIF_SUMMARY	We show that regular measurement of sHER2, CA15.3, and CEA levels is useful for predicting the therapeutic response and for monitoring HER2-targeted therapy in patients with HER2-positive metastatic breast cancer.
Literature	GENERIF_SUMMARY	The value of serum Krebs von den lungen-6 as a diagnostic marker in connective tissue disease associated with interstitial lung disease.
Literature	GENERIF_SUMMARY	MUC1 mediates Pneumocystis murina binding to airway epithelial cells.
Literature	GENERIF_SUMMARY	Breast cancer cutaneous metastases are associated to uMUC1 and sialyl Lewis x and to highly malignant primary tumors.
Literature	GENERIF_SUMMARY	MUC1 oncoprotein mitigates ER stress via CDA-mediated reprogramming of pyrimidine metabolism.

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Literature	GENERIF_SUMMARY	MUC1 is associated with TFF2 methylation in gastric cancer.
Literature	GENERIF_SUMMARY	miR-485-5p inhibits the progression of breast cancer cells by negatively regulating MUC1.
Literature	GENERIF_SUMMARY	Can VEGFR-3 be a better tumor marker for breast cancer than CA 15-3?
Literature	GENERIF_SUMMARY	Altered Expression of the Epithelial Mucin MUC1 Accompanies Endoscopic Recurrence of Postoperative Crohn's Disease.
Literature	GENERIF_SUMMARY	MUC1 promotes glycolysis through inhibiting BRCA1 expression in pancreatic cancer
Literature	GENERIF_SUMMARY	Prognostic significance of EGFR, MUC1 and PD-L1 expressions in cases with triple negative breast cancer.
Literature	GENERIF_SUMMARY	Detection of circulating natural antibodies against CD25, MUC1, and VEGFR1 for early diagnosis of non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Increased MUC1 plus a larger quantity and complex size for MUC5AC in the peripheral airway lumen of long-term tobacco smokers.
Literature	GENERIF_SUMMARY	MUC1 contributes to goblet cell metaplasia and MUC5AC expression in response to cigarette smoke in vivo.
Literature	GENERIF_SUMMARY	Clinical and genetic spectra of autosomal dominant tubulointerstitial kidney disease due to mutations in UMOD and MUC1.
Literature	GENERIF_SUMMARY	SMRT sequencing revealed to be an effective method for ADTKD-MUC1 diagnosis through follow-up analysis of a Chinese family.
Literature	GENERIF_SUMMARY	Serum KL-6 concentrations as a novel biomarker of severe COVID-19.
Literature	GENERIF_SUMMARY	Preclinical evaluation of an (111)In/(225)Ac theranostic targeting transformed MUC1 for triple negative breast cancer.
Literature	GENERIF_SUMMARY	Exploration and correlation analysis of changes in Krebs von den Lungen-6 levels in COVID-19 patients with different types in China.
Literature	GENERIF_SUMMARY	Serum Krebs von den Lungen-6 level in the disease progression and treatment of Mycobacterium avium complex lung disease.
Literature	GENERIF_SUMMARY	Krebs von den Lungen-6 as a biomarker for disease severity assessment in interstitial lung disease
Literature	GENERIF_SUMMARY	A SERS-colorimetric dual-mode aptasensor for the detection of cancer biomarker MUC1.
Literature	GENERIF_SUMMARY	Regulation of mucin 1 expression and its relationship with oral diseases.

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Literature	GENERIF_SUMMARY	Association of MUC1 5640G>A and PSCA 5057C>T polymorphisms with the risk of gastric cancer in Northern Iran.
Literature	GENERIF_SUMMARY	CircHECTD1 up-regulates mucin 1 expression to accelerate hepatocellular carcinoma development by targeting microRNA-485-5p via a competing endogenous RNA mechanism.
Literature	GENERIF_SUMMARY	ISOGlyP
Literature	GENERIF_SUMMARY	O-linked mucin-type glycosylation regulates the transcriptional programme downstream of EGFR.
Literature	GENERIF_SUMMARY	GalNAc-T3 and MUC1, a combined predictor of prognosis and recurrence in solitary pulmonary adenocarcinoma initially diagnosed as malignant solitary pulmonary nodule (
Literature	GENERIF_SUMMARY	Elevated MUC1 and MUC5AC mucin protein levels in airway mucus of critical ill COVID-19 patients.
Literature	GENERIF_SUMMARY	Downregulated Mucin 1 alleviates paclitaxel resistance in nonsmall cell lung cancer cells.
Literature	GENERIF_SUMMARY	Smoking-associated increase in mucins 1 and 4 in human airways.
Literature	GENERIF_SUMMARY	Krebs von den Lungen-6 (KL-6) is a pathophysiological biomarker of early-stage acute hypersensitivity pneumonitis among pigeon fanciers.
Literature	GENERIF_SUMMARY	Dendritic Cell-based Immunotherapy Pulsed With Wilms Tumor 1 Peptide and Mucin 1 as an Adjuvant Therapy for Pancreatic Ductal Adenocarcinoma After Curative Resection
Literature	GENERIF_SUMMARY	TF-containing MUC1 glycopeptides fail to entice Galectin-1 recognition of tumor-associated Thomsen-Freidenreich (TF) antigen (CD176) in solution.
Literature	GENERIF_SUMMARY	Inhibition of MUC1 exerts cell-cycle arrest and telomerase suppression in glioblastoma cells.
Literature	GENERIF_SUMMARY	Polymorphisms PSCA rs2294008, IL-4 rs2243250 and MUC1 rs4072037 are associated with gastric cancer in a high risk population.
Literature	GENERIF_SUMMARY	Associations between the expression of mucins (MUC1, MUC2, MUC5AC and MUC6) and clinicopathologic parameters of human breast carcinomas.
Literature	GENERIF_SUMMARY	Krebs von den Lungen 6 decreased in the serum and muscle of GNE myopathy patients.
Literature	GENERIF_SUMMARY	Could KL-6 levels in COVID-19 help to predict lung disease?
Literature	GENERIF_SUMMARY	C-C motif chemokine ligand 5 and C-C chemokine receptor type 5
Literature	GENERIF_SUMMARY	MUC1-C Activates the BAF (mSWI/SNF) Complex in

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		Prostate Cancer Stem Cells.
Literature	GENERIF_SUMMARY	A Rare Kidney Disease To Cure Them All? Towards Mechanism-Based Therapies for Proteinopathies.
Literature	GENERIF_SUMMARY	In Silico Identification of Potential Druggable Binding Sites on CIN85 SH3 Domain.
Literature	GENERIF_SUMMARY	Racial/ethnic differences in average CA125 and CA15.3 values and its correlates among postmenopausal women in the USA.
Literature	GENERIF_SUMMARY	MUC1-C integrates activation of the IFN-gamma pathway with suppression of the tumor immune microenvironment in triple-negative breast cancer.
Literature	GENERIF_SUMMARY	The Transmembrane Mucin MUC1 Facilitates beta1-Integrin-Mediated Bacterial Invasion.
Literature	GENERIF_SUMMARY	Ultrasensitive Nanopore Sensing of Mucin 1 and Circulating Tumor Cells in Whole Blood of Breast Cancer Patients by Analyte-Triggered Triplex-DNA Release.
Literature	GENERIF_SUMMARY	Krebs Von den Lungen-6 as a predictive indicator for the risk of secondary pulmonary fibrosis and its reversibility in COVID-19 patients.
Literature	GENERIF_SUMMARY	Classification, structural biology, and applications of mucin domain-targeting proteases.
Literature	GENERIF_SUMMARY	Krebs von den Lungen 6 (KL-6) as a marker for disease severity and persistent radiological abnormalities following COVID-19 infection at 12 weeks.
Literature	GENERIF_SUMMARY	Serum Krebs von den Lungen-6 levels are associated with mortality and severity in patients with coronavirus disease 2019.
Literature	GENERIF_SUMMARY	Association between molecular subtype of local advanced breast cancer with Ca 15-3 level.
Literature	GENERIF_SUMMARY	Plasma Mucin-1 (CA15-3) Levels in Autosomal Dominant Tubulointerstitial Kidney Disease due to MUC1 Mutations.
Literature	GENERIF_SUMMARY	Type I Interferon Dependent hsa-miR-145-5p Downregulation Modulates MUC1 and TLR4 Overexpression in Salivary Glands From Sjogren's Syndrome Patients.
Literature	GENERIF_SUMMARY	MUC1-C activates the PBAF chromatin remodeling complex in integrating redox balance with progression of human prostate cancer stem cells.
Literature	GENERIF_SUMMARY	MUC1 is an oncoprotein with a significant role in apoptosis (Review).
Literature	GENERIF_SUMMARY	WWP1 targeting MUC1 for ubiquitin-mediated lysosomal degradation to suppress carcinogenesis.
Literature	GENERIF_SUMMARY	Clinical relevance of serum Krebs von den Lungen-6

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		levels in patients with coronavirus disease 2019.
Literature	GENERIF_SUMMARY	MUC1-C Contributes to the Maintenance of Human Embryonic Stem Cells and Promotes Somatic Cell Reprogramming.
Literature	GENERIF_SUMMARY	sFlt-1 and CA 15.3 are indicators of endothelial damage and pulmonary fibrosis in SARS-CoV-2 infection.
Literature	GENERIF_SUMMARY	Association of Preoperative Serum Levels of CEA and CA15-3 with Molecular Subtypes of Breast Cancer.
Literature	GENERIF_SUMMARY	Pleiotropic Effects of Functional MUC1 Variants on Cardiometabolic, Renal, and Hematological Traits in the Taiwanese Population.
Literature	GENERIF_SUMMARY	A Possible Inhibitory Role of Sialic Acid on MUC1 in Peritoneal Dissemination of Clear Cell-Type Ovarian Cancer Cells.
Literature	GENERIF_SUMMARY	Serum Krebs von den Lungen-6 for Predicting the Severity of COVID-19 Lung Injury
Literature	GENERIF_SUMMARY	MUC1-C dictates neuroendocrine lineage specification in pancreatic ductal adenocarcinomas.
Literature	GENERIF_SUMMARY	Mucin 1 regulates the hypoxia response in head and neck cancer cells.
Literature	GENERIF_SUMMARY	Inhibition of hypoxia-induced Mucin 1 alters the proteomic composition of human osteoblast-produced extracellular matrix, leading to reduced osteogenic and angiogenic potential.
Literature	GENERIF_SUMMARY	The sialidase NEU1 directly interacts with the juxtamembranous segment of the cytoplasmic domain of mucin-1 to inhibit downstream PI3K-Akt signaling.
Literature	GENERIF_SUMMARY	Role of MUC1 rs4072037 polymorphism and serum KL-6 levels in patients with antisynthetase syndrome.
Literature	GENERIF_SUMMARY	Serum Krebs von den Lungen-6 concentrations reflect severity of anti-melanoma differentiation-associated protein 5 antibody positive dermatomyositis associated interstitial lung disease.
Literature	GENERIF_SUMMARY	The human gut symbiont Ruminococcus gnavus shows specificity to blood group A antigen during mucin glycan foraging
Literature	GENERIF_SUMMARY	KL-6 as an Immunological Biomarker Predicts the Severity, Progression, Acute Exacerbation, and Poor Outcomes of Interstitial Lung Disease
Literature	GENERIF_SUMMARY	MUC1 and CD147 Are Promising Markers for the Detection of Circulating Tumor Cells in Small Cell Lung Cancer.
Literature	GENERIF_SUMMARY	Establishment of Patient-derived Preclinical Models for Invasive Papillary Cholangiocarcinoma.
Literature	GENERIF_SUMMARY	MUC1-C Dictates JUN and BAF-Mediated Chromatin

Main Category	Category	Terms
		Remodeling at Enhancer Signatures in Cancer Stem Cells.
Literature	GENERIF_SUMMARY	Expression of Mucin1 in saliva in oral squamous cell carcinoma and oral potentially malignant disorders (case control study).
Literature	GENERIF_SUMMARY	MUC1-C integrates type II interferon and chromatin remodeling pathways in immunosuppression of prostate cancer.
Literature	GENERIF_SUMMARY	A combination of immunohistochemical markers, MUC1, MUC5AC, PAX8 and growth pattern for characterization of mucinous neoplasm of the ovary.
Literature	GENERIF_SUMMARY	MUC1 and Polarity Markers INADL and SCRIB Identify Salivary Ductal Cells.
Literature	GENERIF_SUMMARY	Changes in blood Krebs von den Lungen-6 predict the mortality of patients with acute exacerbation of interstitial lung disease.
Literature	GENERIF_SUMMARY	Expression of Mucins in Different Entities of Salivary Gland Cancer
Literature	GENERIF_SUMMARY	MUC1 triggers lineage plasticity of Her2 positive mammary tumors.
Literature	GENERIF_SUMMARY	Expression of S100A9 and KL-6 in common interstitial lung diseases.
Literature	GENERIF_SUMMARY	Baseline serum Krebs von den Lungen-6 as a biomarker for the disease progression in idiopathic pulmonary fibrosis.
Literature	GENERIF_SUMMARY	Expression Profile of CD10, BCL-2, p63, and EMA in the Normal Skin and Basal Cell Carcinomas
Literature	GENERIF_SUMMARY	The role of epithelial membrane antigen (EMA) overexpression in the prognosis of prostatic adenocarcinoma.
Literature	GENERIF_SUMMARY	Addiction of Merkel cell carcinoma to MUC1-C identifies a potential new target for treatment.
Literature	GENERIF_SUMMARY	Membrane-Tethered Mucin 1 Is Stimulated by Interferon and Virus Infection in Multiple Cell Types and Inhibits Influenza A Virus Infection in Human Airway Epithelium.
Literature	GENERIF_SUMMARY	YKL-40 and KL-6 Levels in Serum and Sputum of Patients Diagnosed With Hypersensitivity Pneumonitis.
Literature	GENERIF_SUMMARY	A prognosis marker MUC1 correlates with metabolism and drug resistance in bladder cancer
Literature	GENERIF_SUMMARY	Prognostic impact of MUC1 and potential regulatory miR-145 and miR-21 expression in salivary mucoepidermoid carcinoma.
Literature	GENERIF_SUMMARY	Effect of smoking on MUC1 expression in oral epithelial dysplasia, oral cancer, and irradiated oral epithelium.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	MUC1
Literature	GENERIF_SUMMARY	The oncoprotein MUC1 facilitates breast cancer progression by promoting Pink1-dependent mitophagy via ATAD3A destabilization.
Literature	GENERIF_SUMMARY	Evaluation of MUC1, MUC2, MUC5AC, and MUC6 Expression Differences in Lung Adenocarcinoma Subtypes by Using a Final Immunoreactivity Score (FIRS).
Literature	GENERIF_SUMMARY	Increased KL-6 levels in moderate to severe COVID-19 infection.
Literature	GENERIF_SUMMARY	MUC1-C Dictates PBRM1-Mediated Chronic Induction of Interferon Signaling, DNA Damage Resistance, and Immunosuppression in Triple-Negative Breast Cancer.
Literature	GENERIF_SUMMARY	The multifaceted role of MUC1 in tumor therapy resistance.
Literature	GENERIF_SUMMARY	MUC1 promotes cancer stemness and predicts poor prognosis in osteosarcoma.
Literature	GENERIF_SUMMARY	Influence of glycoprotein MUC1 on trafficking of the Ca[2+]-selective ion channels, TRPV5 and TRPV6, and on in vivo calcium homeostasis.
Literature	GENERIF_SUMMARY	MUC1-C is a master regulator of MICA/B NKG2D ligand and exosome secretion in human cancer cells.
Literature	GENERIF_SUMMARY	The Glycosylated N-Terminal Domain of MUC1 Is Involved in Chemoresistance by Modulating Drug Permeation Across the Plasma Membrane.
Literature	GENERIF_SUMMARY	Effects of enhancer of zeste homolog 2 and mucin 1 expressions on treatment response in breast cancer.
Literature	GENERIF_SUMMARY	Dependence on MUC1-C in Progression of Neuroendocrine Prostate Cancer.
Literature	GENERIF_SUMMARY	MUC1-C is necessary for SHP2 activation and BRAF inhibitor resistance in BRAF(V600E) mutant colorectal cancer.
Literature	GENERIF_SUMMARY	MUC1 Expression Affects the Immunoflogosis in Renal Cell Carcinoma Microenvironment through Complement System Activation and Immune Infiltrate Modulation.
Literature	GENERIF_SUMMARY	Novel MUC1 variant identified by massively parallel sequencing explains interstitial kidney disease in a large Dutch family.
Literature	GENERIF_SUMMARY	Canonical and truncated transglutaminase-2 regulate mucin-1 expression and androgen independency in prostate cancer cell lines.
Literature	GENERIF_SUMMARY	Serum KL-6 levels predict clinical outcomes and are associated with MUC1 polymorphism in Japanese patients with COVID-19.
Literature	GENERIF_SUMMARY	Hypoxia controls expression of kidney-pathogenic

Main Category	Category	Terms
		MUC1 variants.
Literature	GENERIF_SUMMARY	[Autophagy Markers Induced by Influenza Virus and MUC1 Expressions in Cancer-Derived Cell Lines].
Literature	GENERIF_SUMMARY	MUC1 promotes lymph node metastasis in esophageal squamous cell carcinoma by downregulating DNAJB6 expression.
Literature	GENERIF_SUMMARY	Genome-wide association study meta-analysis supports association between MUC1 and ectopic pregnancy.
Literature	GENERIF_SUMMARY	MUC1 attenuates neutrophilic airway inflammation in asthma by reducing NLRP3 inflammasome-mediated pyroptosis through the inhibition of the TLR4/MyD88/NF-kappaB pathway.
Literature	GENERIF_SUMMARY	Prognostic value of serum Krebs von den Lungen-6 (KL-6) levels in COVID-19 pneumonia.
Literature	GENERIF_SUMMARY	Disease progression and changes in KL-6 in patients with limited cutaneous systemic sclerosis-associated interstitial lung disease.
Literature	GENERIF_SUMMARY	Role of the cAMP-PKA-NF-kappaB pathway in Mucin1 over-expression in A549 cells during Respiratory syncytial virus infection.
Literature	GENERIF_SUMMARY	Lower myostatin and higher MUC1 levels are associated with better response to mepolizumab and omalizumab in asthma
Literature	GENERIF_SUMMARY	Endometrial receptivity defects MUC-1 and COX-2 polymorphisms in endometriosis.
Literature	GENERIF_SUMMARY	Eliminating the concept of unknown chronic kidney disease
Literature	GENERIF_SUMMARY	Mucin 1 and venous thrombosis in tumor-bearing mice and patients with cancer.
Literature	GENERIF_SUMMARY	Identification of MUC1-C as a Target for Suppressing Progression of Head and Neck Squamous Cell Carcinomas.
Literature	GENERIF_SUMMARY	Evaluation of mucin-1, nuclear factor kappaB, and hemoglobin A1c levels in obese and non-obese individuals.
Literature	GENERIF_SUMMARY	XIST and MUC1-C form an auto-regulatory pathway in driving cancer progression.
Literature	GENERIF_SUMMARY	NSUN2 affects diabetic retinopathy progression by regulating MUC1 expression through RNA m[5]C methylation.
Literature	GENERIF_SUMMARY	Serum YKL-40 and Serum Krebs von den Lungen-6 as Potential Predictive Biomarkers for Rheumatoid Arthritis-Associated Interstitial Lung Disease.
Literature	GENERIF_SUMMARY	TRPV4 activation in human corneal epithelial cells

Main Category	Category	Terms
		promotes membrane mucin production.
Pathways	REACTOME_PATHWAY	Developmental Biology
Pathways	REACTOME_PATHWAY	Cytokine Signaling in Immune system
Pathways	REACTOME_PATHWAY	Disease
Pathways	REACTOME_PATHWAY	Innate Immune System
Pathways	REACTOME_PATHWAY	Immune System
Pathways	REACTOME_PATHWAY	Diseases of glycosylation
Pathways	REACTOME_PATHWAY	Diseases associated with O-glycosylation of proteins
Pathways	REACTOME_PATHWAY	Metabolism of proteins
Pathways	REACTOME_PATHWAY	Signaling by Interleukins
Pathways	REACTOME_PATHWAY	Defective GALNT3 causes HFTC
Pathways	REACTOME_PATHWAY	Defective C1GALT1C1 causes TNPS
Pathways	REACTOME_PATHWAY	Defective GALNT12 causes CRCS1
Pathways	REACTOME_PATHWAY	O-linked glycosylation
Pathways	REACTOME_PATHWAY	Dectin-2 family
Pathways	REACTOME_PATHWAY	C-type lectin receptors (CLRs)
Pathways	REACTOME_PATHWAY	Diseases of metabolism
Pathways	REACTOME_PATHWAY	Post-translational protein modification
Pathways	REACTOME_PATHWAY	Interleukin-4 and Interleukin-13 signaling
Pathways	REACTOME_PATHWAY	O-linked glycosylation of mucins
Pathways	REACTOME_PATHWAY	Developmental Cell Lineages
Pathways	REACTOME_PATHWAY	Developmental Cell Lineages of the Integumentary System
Pathways	REACTOME_PATHWAY	Termination of O-glycan biosynthesis
Pathways	REACTOME_PATHWAY	Developmental Lineages of the Mammary Gland
Pathways	REACTOME_PATHWAY	Developmental Lineage of Mammary Gland Luminal Epithelial Cells
Pathways	REACTOME_PATHWAY	Developmental Lineage of Mammary Gland Alveolar Cells
Pathways	WIKIPATHWAYS	Hematopoietic stem cell differentiation
Pathways	WIKIPATHWAYS	Altered glycosylation of MUC1 in tumor microenvironment
Protein Domains	INTERPRO	SEA_dom

Main Category	Category	Terms
Protein Domains	INTERPRO	SEA_dom_sf
Protein Domains	SMART	SEA
Protein Domains	UP_KW_DOMAIN	Repeat
Protein Domains	UP_KW_DOMAIN	Signal
Protein Domains	UP_KW_DOMAIN	Transmembrane
Protein Domains	UP_KW_DOMAIN	Transmembrane helix